The Effect of Class Imbalance on Causal Effect Estimation

by
Abel Mesfin

Professor Rohit Bhattacharya, Advisor

A thesis submitted in partial fulfillment
of the requirements for the
Degree of Bachelor of Arts with Honors
in Computer Science

Williams College
Williamstown, Massachusetts

May, 2026

Contents

1 Introduction and Motivation **7**
- 1.1 Motivating Example . . . 8
- 1.2 Revealing the Oracle . . . 9
- 1.3 Takeaways . . . 10
- 1.4 Related Work . . . 11

2 Background **13**
- 2.1 Statistical DAG Models . . . 13
- 2.2 Causal DAG Models . . . 15
- 2.3 d-Separation . . . 16
- 2.4 Recovering Causal Effects . . . 18
  - 2.4.1 Backdoor Adjustment . . . 18
  - 2.4.2 Inverse Probability Weighted Estimator (IPW) . . . 18
  - 2.4.3 Augmented Inverse Probability Weighting (AIPW) . . . 19
- 2.5 Selection Bias . . . 20
  - 2.5.1 Recoverability in Markovian Models . . . 22
- 2.6 Machine Learning . . . 23
- 2.7 Logistic Regression . . . 24
- 2.8 Neural Networks . . . 25
- 2.9 Class Imbalance . . . 26
  - 2.9.1 Data-level techniques . . . 26
  - 2.9.2 Algorithm-level techniques . . . 27
  - 2.9.3 Hybrid Approaches . . . 28
- 2.10 Evaluation Metrics for Imbalanced Classification . . . 28

3 SMOTE **30**
- 3.1 How does SMOTE work? . . . 30
  - 3.1.1 Algorithm . . . 31
- 3.2 Causal Implications of SMOTE . . . 32
- 3.3 Distribution Shift after Applying SMOTE . . . 32
- 3.4 Simulation Counterexample for SMOTE . . . 34

4 New reweighted AIPW Estimator **35**
- 4.1 Causal View of Resampling Methods . . . 35
- 4.2 Reweighted AIPW Estimator for Rare Treatment . . . 37
  - 4.2.1 Proof of rAIPW for rare treatment . . . 38
  - 4.2.2 Steps to Computing ACE using rAIPW in rare treatment . . . 39
- 4.3 Reweighted AIPW Estimator for Rare Outcome . . . 39
  - 4.3.1 Proof of rAIPW for rare outcome . . . 40

4.3.2 Steps to Computing ACE using rAIPW in rare outcomes 41
4.4 Recovering Propensity Scores with the Odds Ratio Factorization 41

5 Simulation Study **44**
5.1 Data-Generating Process 44
5.2 Estimators and Evaluation Protocol 45
5.3 Results 45
5.3.1 Point Estimates Under Fixed Sample Size 45
5.3.2 Convergence Across Sample Sizes 46
5.4 Summary 47

6 Discussion and Conclusion **49**
6.1 Recommendations for Class Imbalance 49
6.2 Limitations 50
6.3 Future Work 50

A Additional Simulation Details **52**
A.1 Additional Details regarding Simulations in Section 1.1 52
A.2 Details regarding Simulations in Section 1.2 52

List of Figures

1.1 Oracle causal graph for motivating example. 9
1.2 Causal graph that shows optimizing over outcome regression can lead to a biased estimate. 12

2.1 The three fundamental DAG triplets. 16
2.2 A collider ($A \to B \leftarrow C$) with descendant D. Conditioning on D opens the path between A and C. 17
2.3 Berkson's bias through clinic-based selection. 21
2.4 Outcome-dependent selection in the job training example. 21
2.5 Recoverability succeeds since $Y \perp\!\!\!\perp R_S \mid A$. 23
2.6 Recoverability fails since $Y \not\perp\!\!\!\perp R_S \mid A$. 23

3.1 Causal diagram for SMOTE under treatment imbalance. Synthetic covariates Z_S are generated by interpolating from both the real observation and its nearest neighbor. The synthetic treatment label A_S is assigned deterministically, so no edge from A_{nn} to A_S exists. 32
3.2 Causal diagram for the data-generating process. Z is a confounder affecting both treatment A and outcome Y. Conditioning on Z satisfies the backdoor criterion. . . 34

4.1 Causal diagram illustrating outcome-dependent resampling 35
4.2 Causal diagram illustrating treatment-dependent resampling 36
4.3 Causal diagram illustrating class-weighting 37

5.1 Causal diagram for the simulation study. Z confounds the effect of treatment A on outcome Y. 44
5.2 Distribution of ACE estimates across 200 simulation replicates per sample size under **undersampling**. Left: standard AIPW applied to the balanced subset. Right: rAIPW applied to the full dataset with resampling weights. The dashed line marks the true ACE = 1.9. 47
5.3 Distribution of ACE estimates across 200 simulation replicates per sample size under **oversampling**. Left: standard AIPW applied to the balanced dataset. Right: rAIPW applied to the full dataset with resampling weights. The dashed line marks the true ACE = 1.9. 47

List of Tables

5.1 Point estimates and 95% bootstrap confidence intervals for the ACE under four resampling strategies, two estimators, and two nuisance model classes. Fixed sample size $n = 2{,}000$; true ACE = 1.9. 46

Abstract

Canonical machine learning practice for handling class imbalance—including undersampling, oversampling, synthetic minority oversampling technique (SMOTE), and class weighting—was developed with predictive precision as its primary objective. However, when these techniques are applied during nuisance model estimation in a causal inference pipeline, they can compromise the validity of causal effect estimates. This thesis investigates the theoretical and empirical consequences of applying common class imbalance remedies in observational causal inference settings.

We formalize the problem using the framework of causal directed acyclic graphs (DAGs) and the recoverability theory of Bareinboim and Tian (2015). We show that each standard resampling strategy—whether treatment-dependent or outcome-dependent—introduces an implicit selection mechanism that violates the conditional independence assumption required for causal recoverability. For SMOTE, we prove that synthetic interpolation shifts the marginal distribution of covariates, leading to biased estimates of the true causal estimand.

To address these failures, we propose a novel reweighted Augmented Inverse Probability Weighting (rAIPW) estimator that corrects for the selection bias introduced by resampling. We derive the estimator and provide formal proofs under both treatment-dependent and outcome-dependent resampling. We also show how the propensity score can be recovered from the resampled data via the odds-ratio factorization of Chen (2007). We validate our theoretical findings through a simulation study in which the true average causal effect (ACE) is known by construction. The results confirm that applying standard AIPW after resampling yields biased estimates, whereas the proposed rAIPW estimator reliably recovers the true ACE across all resampling strategies.

Acknowledgments

Glory be to the Most High God forever, who by His grace has helped me from the beginning unto the end.

Next, I owe my deepest gratitude to my thesis advisor, Professor Rohit Bhattacharya, for his guidance throughout my journey at Williams College. Thank you for being not only my thesis advisor but also my mentor through the Allison Davis Research Program over the past two years. This thesis would not have been possible without your support, encouragement, and belief in me. I will be forever grateful to all the causes that brought me to your CSCI 134 class. I would also like to thank my second reader, Prof. Aaron Williams, for the thoughtful comments and engagement with this work.

I am also sincerely grateful to Robert Blay, Ivonne Garcia, Clinton Williams, and the entire leadership team of the Mellon Mays Undergraduate Fellowship and the Allison Davis Research Program. Thank you for welcoming me with open arms. I have grown tremendously since the sophomore summer when I joined this community, which has continually pushed me beyond my comfort zone and helped me become a more confident and independent researcher.

My beloved family, your prayers and love have carried me through every stage of this journey. Mother, your quiet sacrifices made this possible in ways I'll never fully repay. Thomas, having you as a brother has been one of the greatest gifts of my life. And Grandma, you are the foundation on which everything I've built stands. May God grant you long life, health, and peace.

I am also deeply grateful to Zaid, Abraham, and all of my family in Atlanta—thank you for being my home away from home.

To my friends—Sarah, Aike, Bisola, Max, and Isaac—thank you for always being there for me and for putting up with me through everything. And thank you, Simon, for dragging me to the squash court and somehow convincing me to exercise once in a blue moon.

Chapter 1

Introduction and Motivation

Computing the effects of interventions is one of the central challenges in the empirical sciences. Across disciplines, data are collected with the goal of understanding how deliberate changes to a system would alter its behavior: What is the impact of a new taxation policy? How should a robot respond to unexpected events? Will an advertisement campaign increase users' purchasing behavior? How effective is a new drug for treating cancer? These are fundamentally causal questions.

Causal inference provides a rigorous framework for answering such questions by determining how outcomes would change under alternative actions or interventions. Unlike traditional statistical analyses, which describe associations, causal inference seeks to estimate cause-and-effect relationships. This requires reasoning about counterfactual or potential outcome worlds: what would have happened if a different treatment or action had been applied.

A major difficulty in computing causal effects is the presence of *confounding*. Randomized controlled trials (RCTs), the gold standard for establishing causal effects, eliminate confounding by ensuring that treatment assignment is statistically independent of all factors—measured or unmeasured—that influence the outcome. However, RCTs are often infeasible due to ethical, financial, or technical constraints. As a result, researchers must rely on observational data, where individuals follow their natural behaviors rather than being randomly assigned.

To illustrate these challenges concretely, consider the question of whether vaping increases the risk of lung cancer. In principle, an ideal RCT would randomly assign some individuals to vape and others not to vape, ensuring comparability between groups. Under such a design, the causal effect could be identified by directly comparing average cancer incidence between the randomized groups.

In reality, such a trial is ethically impossible: one cannot deliberately assign individuals to engage in a potentially harmful behavior. Instead, researchers must analyze observational data in which individuals self-select into vaping or non-vaping groups. This self-selection introduces systematic differences between groups—such as age, stress level, risk preferences, or socioeconomic status—that also influence lung cancer risk. These confounding factors obscure the true effect of vaping and highlight the central difficulty of causal inference.

This difficulty is formalized in the *fundamental problem of causal inference*: for any individual, we can observe only one potential outcome—the outcome under the treatment they actually re-

ceived—but never the outcome they would have experienced under the alternative treatment. Let Y^{a_1} denote the outcome an individual would have if they vape, and Y^{a_0} the outcome if they do not. The individual-level causal effect, $Y^{a_1} - Y^{a_0}$, is therefore unobservable. Consequently, causal inference focuses on population-level quantities such as the *Average Causal Effect* (ACE),

$$ACE = \mathbb{E}[Y^{a_1}] - \mathbb{E}[Y^{a_0}],$$

which captures the expected change in risk under a hypothetical intervention that assigns the entire population to vape or not vape. Estimating the ACE from observational data requires assumptions about how the data were generated; if these assumptions fail, estimates of the causal effect become biased or misleading.

Causal inference revolves around two key components: *identification* and *estimation*. Identification determines whether a causal effect can be uniquely expressed in terms of observable quantities under a set of structural assumptions. Estimation then applies statistical or machine-learning methods to approximate that effect once it is identified. Together, these steps link assumptions about the data-generating process to the mathematical procedures used to quantify the consequences of interventions.

While causal identification provides the theoretical basis for estimating causal effects, the quality of estimation in practice depends heavily on the structure of the data. In many modern applications—including healthcare, public policy, and digital platforms—observational datasets are highly imbalanced. Some treatment groups or outcomes occur far less frequently than others, which can lead to unstable estimates, poor overlap, and large variance in causal effect estimates. In binary settings, the class with many instances is called the majority class, while the one with few instances is the minority class. This scenario is common in real-world applications where detecting rare events is crucial: fraudulent transactions, rare disease diagnosis, natural disaster prediction, and software defect detection.

Most standard classification algorithms implicitly assume balanced class distributions and optimize for overall accuracy. When applied to imbalanced data, however, these models tend to favor the majority class, often achieving deceptively high accuracy while failing to detect the rare—but often most consequential—instances. In high-stakes settings such as medicine, overlooking a rare but severe condition can be far more harmful than misclassifying a common one. As a result, overall accuracy becomes a misleading evaluation metric. Alternative measures such as precision, recall, sensitivity, specificity, and the F1 score provide a more informative assessment of model performance under imbalance. We will examine these metrics in detail in the following chapter.

1.1 Motivating Example

To motivate our main research question, we present an example a data analyst would encounter at true inference time, and then we reveal the true DGP and causal effects in §1.2. In this example, the covariates Z are assumed (correctly) to be a sufficient backdoor adjustment set, i.e., a set of variables that are sufficient to adjust for all sources of confounding. Full details of the data-generating process

for the example is provided in the appendix.

The dataset consists of 5000 observations, of which only 21.9% receive treatment, resulting in class imbalance. To estimate the propensity score, the analyst fits logistic regression models for $p(A \mid O)$, where $O \subset Z$ is a subset of the most important confounding variables, both with and without class-weighted loss functions. These propensity score estimates are then used in inverse probability weighting (IPW) with cross-fitting to estimate the average causal effect (ACE), denoted $\hat{\tau}$. Relevant details for IPW and propensity scores are covered later in the thesis; for now, IPW can be treated as a procedure that produces estimates of the causal effect by reweighting individuals by the inverse of their probability of receiving treatment to adjust for confounding. The analyst evaluates the propensity score models using F1 score and compares the resulting causal effect estimates:

$\hat{\pi}$	F1	$\hat{\tau}$	95% CI
Baseline: Class-weighted $p(A)$	–	1.02	[0.85, 1.21]
Unweighted $p(A \mid O)$	0.07	1.94	[1.82, 2.05]
Class-weighted $p(A \mid O)$	0.43	0.82	[0.66, 0.96]

Guided by standard machine learning practice, the analyst selects the class-weighted $p(A \mid O)$ model, as it achieves a much higher F1 score than the unweighted alternative, and reports the corresponding estimate of $\hat{\tau}$. This choice appears well-justified from a predictive modeling perspective. Then the natural question that follows is, do best practices from machine learning always correspond to the "correct" decisions for causal estimation? In other words, are these examples "just" functional estimation (supervised ML) problems in isolation of causal identification? We argue the answer to these questions is "NO" by revealing the oracle true causal effects in the next section.

1.2 Revealing the Oracle

The data for motivating example is generated from the causal graph in Figure 1.1, where A denotes a binary treatment, Y the outcome of interest, and $Z = \{W_1, W_2, O_1, O_2, O_3\}$ the set of observed covariates. Given the causal graph, causal identification is established—adjusting for $W = \{W_1, W_2\}$, $\emptyset = \{O_1, O_2, O_3\}$, or the full set Z blocks all backdoor (i.e., non-causal) paths from A to Y. See the full DGP in the appendix A.2.

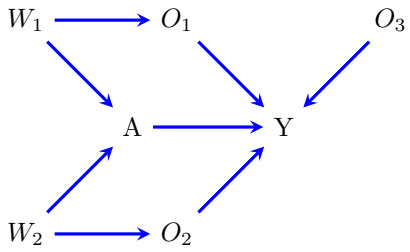


Figure 1.1: Oracle causal graph for motivating example.

The true ACE is 2.0. Counterintuitively, using class weighting for the propensity score model results in an increased F1 score for the nuisance model but a **biased** causal effect estimate whose

confidence interval does not cover the true ACE. Instead, a propensity score model with a low F1 score results in a causal effect estimate which is close to the true ACE (minus some finite-sample bias) and has a small confidence interval which covers the true ACE. In this thesis, we elaborate on the theoretical reasons for this causal bias. In addition to class weighting, we look at other resampling techniques such as undersampling, oversampling and a more sophisticated technique called the SMOTE. We delve into these techniques in the later chapters.

1.3 Takeaways

From the analyst's perspective at inference time, the true causal graph and the true ACE are unknown. Therefore, it is impossible to determine whether a modeling choice improves causal estimation based solely on predictive metrics such as accuracy or F1 score. Unlike supervised learning tasks, causal effect estimates generally cannot be validated against ground truth in observational settings. Unless randomized experiments are available—such as in benchmark evaluation frameworks that use RCT data (e.g., Keith et al. (2023))—there is no direct way to verify whether an estimated causal effect is correct.

The importance of class imbalance began receiving serious attention in the early 2000s, leading to the development of several strategies for handling skewed class distributions. Broadly, these approaches fall into three categories. *Data-level methods* alter the observed distribution by reducing the number of majority-class examples, increasing the number of minority examples, or creating synthetic ones using approaches such as the synthetic minority oversampling technique (SMOTE) (Chawla et al., 2002). *Algorithm-level methods* keep the data intact but modify the learning objective, for instance by assigning higher weights to errors on minority-class observations. *Hybrid methods* draw on both perspectives, combining adjustments to the data with changes to the learning procedure. These techniques were mainly developed with predictive accuracy, not causal identifiablity, as the primary objective. Predictive modeling aims to minimize a loss function, whereas causal inference aims to preserve the structural relationships among treatment, covariates, and outcomes. When undersampling, oversampling, SMOTE, or class weighting are applied in causal settings, they may change the joint distribution in ways that violate the assumptions required for identification. These distortions threaten recoverability of the causal estimand and raise a fundamental questions:

- Can methods designed to improve prediction under class imbalance be safely applied in causal inference without compromising identifiability?
- If identifiability is compromised, can we design a principled reweighting strategy that improves class balance while preserving causal structure?

This work highlights how common statistical and machine learning practices for dealing with real-world challenges such as class imbalance can, in fact, be counterproductive for the purposes of causal effect estimation. In particular, we show that steps in functional estimation can sometimes introduce bias in causal effect estimates by effectively "undoing" the prior step of causal identification. We ground our arguments in well-known results from the literature on causal graphs and

provide empirical evidence of our claims through simulations, and ultimately synthesize these results into concrete recommendations for handling the bias-variance tradeoff in functional estimation while being mindful of causal identification.

1.4 Related Work

Methods for causal effect estimation under rare treatment assignment and rare outcomes remain relatively underexplored. In the rare-outcome setting, Balzer et al. (2016) proposed a bounded targeted minimum loss-based estimator (rTMLE) that incorporates prior knowledge that the outcome regression, $\mu(A, W) = \mathbb{E}[Y \mid A, W]$, should remain small across treatment–covariate strata. By constraining the conditional risk to lie within a realistic upper bound, their approach stabilizes nuisance estimation, reduces finite-sample variance, and improves statistical power while preserving the double robustness guarantees of standard TMLE. Their work highlights how structural knowledge about rare events can improve causal effect estimation in high-dimensional settings.

Our setting is closely related, as we also study causal effect estimation in the presence of rare outcomes. However, in addition to outcome rarity, we also consider treatment imbalance and the widespread use of resampling-based solutions, such as undersampling, oversampling, and class weighting, during nuisance estimation. These procedures can change the effective treatment mechanism or the joint covariate distribution, leading to selection bias or distributional shifts that standard doubly robust estimators do not address. Whereas Balzer et al. (2016) focuses on stabilizing the outcome regression under rare outcomes, our work focuses on recovering valid causal effect estimates after nuisance models are trained on resampled data.

This research examines the tension between predictive accuracy of nuisance models and causal modeling objectives. Variable selection for high-dimensional covariates is often guided by maximizing the predictive performance of nuisance models, but optimizing predictive fit does not guarantee that the selected variables are valid confounders—they may instead be colliders or descendants of the treatment, whose inclusion opens non-causal paths and biases the effect estimate. Tamarchenko (2023) formalizes this tension, showing that selecting variables by optimizing the outcome regression or propensity score in the presence of unmeasured confounding can lead to biased causal estimates. An example of this is shown in Figure 1.2. Because of the colliders on C_1 and C_2 and the local Markov property, $\{C_1, C_2, O_1\}$ will be selected as the adjustment set, which leads to a biased causal estimate due to the lack of d-separation through the C_1-to-C_2 path. The optimal adjustment set is simply $\{\emptyset\}$—A and Y are already d-separated. If collider bias is present in the m-graph—a mixed graph admitting both directed edges representing direct causal effects and bidirected edges representing unmeasured common causes—the resulting causal effect estimate can be biased, and thus optimizing over the outcome regression should be avoided.

Together, these results paint a picture in which the most natural machine learning heuristics—optimize a loss, maximize a performance metric, balance a class distribution—can undermine the identification assumptions on which causal estimates depend.

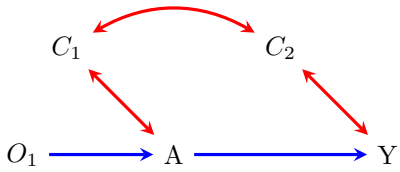


Figure 1.2: Causal graph that shows optimizing over outcome regression can lead to a biased estimate.

Chapter 2

Background

The target parameter, $\mathbb{E}[Y^a]$, cannot be expressed as a function of the observed data, or in other words, is not identified, if no assumptions are made about the data generating process (Pearl, 2009). Graphical models provide a principled way to encode the assumptions needed for identification. In this chapter, we use directed acyclic graphs (DAGs) to formalize the structure of the data-generating process underlying observational studies.

2.1 Statistical DAG Models

A *directed acyclic graph* (DAG) consists of a set of vertices V and directed edges E. An edge points from one vertex to another, and the direction matters (e.g., $A \to Y$ is distinct from $Y \to A$). A *directed path* is a sequence of distinct vertices $V_1, V_2, \ldots, V_k$ such that each pair satisfies $V_i \to V_{i+1}$. A general *path* may include edges of either direction, so each pair satisfies either $V_i \to V_{i+1}$ or $V_{i+1} \to V_i$.

A *directed cycle* is a sequence of directed edges that starts and ends at the same vertex,

$$V_1 \to V_2 \to \cdots \to V_k = V_1.$$

A DAG is defined by the absence of such cycles, ensuring one cannot follow directed edges and return to the starting point. For any vertex V_i, the set of *parents*, denoted $\text{pa}_G(V_i)$, consists of all vertices with arrows pointing into V_i. The *children* of V_i are the vertices receiving arrows from V_i. The *ancestors* of V, denoted $\text{an}_G(V_i)$, is the set of vertices with directed paths into V, which by convention includes V_i itself.

A statistical model associated with a DAG includes a set of random variables V and a joint distribution $p(V)$ that respects the conditional independence relations implied by the graph. For sets of variables A and B, the notation $p(A \mid B)$ denotes the conditional distribution of A given B. If $B = \emptyset$, then $p(A)$ is the marginal distribution.

Any joint distribution can be factorized by the chain rule of probability:

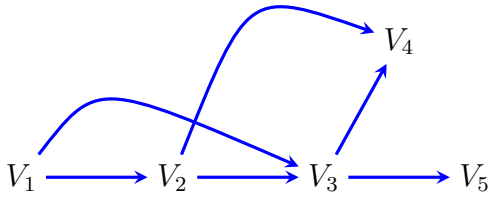
$$p(V_1, V_2, \ldots, V_n) = p(V_1)\, p(V_2 \mid V_1)\, p(V_3 \mid V_1, V_2) \cdots p(V_n \mid V_1, \ldots, V_{n-1}).$$

However, a DAG imposes a more structured factorization. Given a DAG G, its *statistical model* is defined by the recursive decomposition

$$p(V) = \prod_{V_i \in V} p(V_i \mid \mathrm{pa}_G(V_i)). \tag{2.1}$$

We refer to equation (2.1) as the *factorization* of the DAG. This factorization describes precisely how the joint distribution behaves if the data are generated according to the DAG and makes the implied conditional independence relations explicit.

To illustrate, consider observed data generated from the following DAG:



The DAG factorization is

$$p(V_1)\, p(V_2 \mid V_1)\, p(V_3 \mid V_1, V_2)\, p(V_4 \mid V_2, V_3)\, p(V_5 \mid V_3).$$

In contrast, the chain-rule factorization of the full joint distribution is

$$p(V_1)\, p(V_2 \mid V_1)\, p(V_3 \mid V_1, V_2)\, p(V_4 \mid V_1, V_2, V_3)\, p(V_5 \mid V_1, V_2, V_3, V_4).$$

Comparing these expressions reveals the conditional independencies encoded by the DAG. For instance,

$$p(V_4 \mid V_1, V_2, V_3) = p(V_4 \mid V_2, V_3), \qquad p(V_5 \mid V_1, V_2, V_3, V_4) = p(V_5 \mid V_3).$$

Thus V_4 is conditionally independent of V_1 given its parents V_2 and V_3, and V_5 is conditionally independent of V_1, V_2, and V_4 given its parent V_3.

In general, for variables X, Y, and a set Z, the notation

$$X \perp\!\!\!\perp Y \mid Z$$

means that once Z is observed, learning the value of Y provides no additional information about X, and vice versa. DAG factorizations make these relations transparent and form the foundation for identification and causal reasoning in graphical models.

2.2 Causal DAG Models

Causal inference studies the discovery, identification, and estimation of causal effects from data. To articulate causal assumptions formally, researchers often rely on the *Structural Causal Model (SCM)* framework, which is grounded in the theory of *Structural Equation Models (SEMs)* introduced by Pearl (2009). SEMs provide a mathematical representation of causal mechanisms and are closely associated with Directed Acyclic Graphs (DAGs), which visually encode these relationships.

One of the most common ways to endow a DAG with causal meaning is to associate it with a collection of structural equations that describe how each variable in the system is generated. More formally, a Structural Equation Model (SEM) $\mathcal{M}$ corresponding to a DAG $G(V, E)$ consists of equations of the form

$$V_i = f_i(\mathrm{pa}_G(V_i), \varepsilon_i), \qquad \forall V_i \in V,$$

where:

- V_i is an *endogenous variable* whose value is determined by the model,
- $\mathrm{pa}_G(V_i)$ denotes the set of *parent variables* (i.e., direct causes) of V_i in G,
- ε_i is an *exogenous noise term* capturing all factors outside the model,
- f_i is an arbitrary function mapping parents and noise to the value of V_i.

The exogenous errors ε_i are assumed to be mutually independent:

$$p(\varepsilon) = \prod_{i=1}^{n} p(\varepsilon_i).$$

This independence ensures that each structural equation represents an autonomous causal mechanism. Together, the DAG and its structural equations define a coherent generative model: each variable is produced as a function of its direct causes and independent noise. The acyclicity of the graph guarantees that the variables can be evaluated in a logical causal order, typically corresponding to any topological ordering of the DAG.

Interventions in an SEM $\mathcal{M}$ are formalized using the *do-operator* (Pearl, 2009). To intervene on a variable V_i and set its value to v_i, one replaces its structural equation

$$V_i = f_i(\mathrm{pa}_G(V_i), \varepsilon_i)$$

with the deterministic assignment $V_i = v_i$. All downstream variables whose equations depend on V_i are then evaluated at this interventional value. That is, for any V_j such that $V_i \in \mathrm{pa}_G(V_j)$, the structural equation becomes

$$V_j = f_j(\mathrm{pa}_G(V_j), \varepsilon_j)\,\big|_{V_i = v_i}.$$

The notation $|_{V_i=v_i}$ means that every occurrence of V_i inside f_j is replaced by the intervention value v_i.

There is a well-established correspondence between this graphical intervention framework and the potential outcomes notation used in the introduction (see Richardson and Robins (2013); Malinsky et al. (2019)). In particular, the value of V_j in the modified system created by the intervention $\text{do}(V_i = v_i)$ can be interpreted as the potential outcome $V_j^{v_i}$. This equivalence allows us to use the potential outcomes notation while benefiting from the transparency and structure of graphical models.

The formalism above yields a clear interpretation of edges in a causal DAG. A directed edge $A \to B$ indicates that A is a *potential direct cause* of B relative to the other variables in the graph. Conversely, the absence of an edge $A \to B$ asserts that A is *not* a direct cause of B in the context of G. Notably, a missing edge represents a particularly strong assumption: it excludes any direct causal influence from A to B. The presence of an edge, in contrast, indicates only the possibility of a direct effect, not a guarantee.

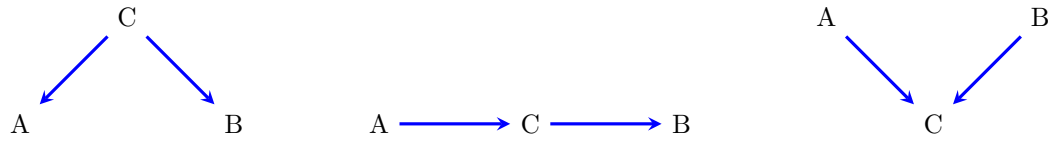
Beyond providing a causal generative story, the structure of a DAG also entails statistical implications. In particular, missing edges encode conditional independence relations and other equality constraints that the observed data distribution must satisfy if the DAG is correct. In the next section, we present *d-separation*, a graphical criterion that allows us to read these implied conditional independencies directly from the graph.

2.3 d-Separation

d-separation is the central graphical criterion used to determine which conditional independence relations are implied by a DAG (Pearl, 2009). Since conditional independence constraints are a key link between causal assumptions and observable statistical structure, we begin by introducing the basic graphical components that underlie the definition of d-separation: paths, forks, chains, and colliders.

A *path* between two vertices V_1 and V_k in a DAG G is a sequence of vertices $V_1, V_2, \ldots, V_k$ such that each consecutive pair is connected by an edge in either direction (i.e., either $V_i \to V_{i+1}$ or $V_{i+1} \to V_i$). A *directed path* is a path in which all arrows point forward, away from the starting variable. A vertex V_j is called a *descendant* of V_i if there exists a directed path from V_i to V_j.

To understand how conditioning affects dependence, we focus on three fundamental three-node structures (triplets), illustrated in Figure 2.1: forks, chains, and colliders.



(a) Fork (b) Chain (c) Collider

Figure 2.1: The three fundamental DAG triplets.

- **Forks ($A \leftarrow C \to B$):** C is a common cause of A and B. The path is *open* unless we condition on C, in which case it becomes *blocked*.

- **Chains ($A \to C \to B$):** C lies on the causal pathway from A to B. As with forks, the path is open unless we condition on C, which blocks it. Chains and forks have identical statistical implications but encode different causal structures.
- **Colliders ($A \to C \leftarrow B$):** C is a common effect of A and B. Unlike forks and chains, the path through a collider is *blocked* unless we condition on C *or any descendant of* C, which opens the path.

Because conditioning affects colliders differently, we must also consider descendants of colliders. Figure 2.2 shows a collider $A \to B \leftarrow C$ with a descendant D of the collider node B. Conditioning on D opens the collider path, thereby inducing dependence between A and C.

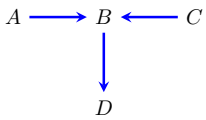


Figure 2.2: A collider ($A \to B \leftarrow C$) with descendant D. Conditioning on D opens the path between A and C.

We are now prepared to formalize when a set of variables blocks a path. Let X and Y be two vertices in V, and Z be any subset of the remaining variables. A path between X and Y is *blocked* by Z if it contains either:

- a fork or chain in which the middle variable is in Z, or
- a collider for which neither the collider nor any of its descendants is in Z.

Definition (D-separation). Two variables X and Y are *d-separated* by a set Z if *every* path between them is blocked by Z. If at least one path remains unblocked, then X and Y are *d-connected* given Z.

The importance of d-separation stems from its relationship to statistical independence. The *global Markov property* states that in any distribution $p(V)$ generated by a DAG G (via its structural equation model $\mathcal{M}$), d-separation implies conditional independence:

$$X \text{ is d-separated from } Y \text{ by } Z \text{ in } G \quad \Longrightarrow \quad X \perp\!\!\!\perp Y \mid Z \text{ in } p(V).$$

To enable testing of causal models from data, we additionally assume the converse implication holds. This is the *faithfulness assumption*:

$$X \text{ is d-separated from } Y \text{ by } Z \quad \Longleftrightarrow \quad X \perp\!\!\!\perp Y \mid Z.$$

Faithfulness rules out pathological cases in which variables appear independent due to finely tuned cancellations among causal pathways rather than due to the structure of the DAG. Although such violations can occur, they require extremely specific parameter values and are considered statistically rare (Spirtes et al., 2001). Under faithfulness, each observed independence corresponds to a structural restriction in the DAG, enabling both causal discovery and valid model checking.

2.4 Recovering Causal Effects

Having established how d-separation links the structure of a DAG to statistical independencies in the observed data, we now turn to the problem of identifying causal effects. In particular, we show how these graphical tools allow us to formalize and compute the Average Causal Effect (ACE).

2.4.1 Backdoor Adjustment

Given a causal graph G, a set of variables Z is said to satisfy the backdoor criterion with respect to a treatment A and outcome Y if

(a) Z does not include any descendants of A

(b) A and Y are d-separated given Z in the graph $G_{\underline{A}}$, which is the same as the original graph G except all outgoing edges $A \to \circ$ have been deleted

When Z satisfies the backdoor criterion with respect to A and Y, the average causal effect is identified via the backdoor adjustment formula (Pearl, 1995),

$$\mathbb{E}[Y^{a_1}] - \mathbb{E}[Y^{a_0}] = \sum_Z p(Z)\,\mathbb{E}[Y \mid a_1, Z] \;-\; \sum_Z p(Z)\,\mathbb{E}[Y \mid a_0, Z].$$

2.4.2 Inverse Probability Weighted Estimator (IPW)

A key difficulty in estimating causal effects from observational data is that the treatment A is often *imbalanced*: some covariate Z make treatment far more or less likely than others. As a result, treated and control units may not be comparable, and naive averages such as $\mathbb{E}[Y \mid A = a]$ generally do *not* identify the counterfactual quantity $\mathbb{E}[Y^a]$. To correct this imbalance, we can reweight the sample so that treatment assignment behaves "as if randomized" within levels of Z.

An alternate method for estimating $\mathbb{E}[Y^a]$ using a valid backdoor adjustment set Z is the inverse probability weighting (IPW) estimator, given by equation 2.8 in Horvitz and Thompson (1952):

$$\mathbb{E}[Y^a] = \mathbb{E}\left[\frac{\mathbb{I}(A = a)\,Y}{p(A = a \mid Z)}\right] \tag{2.8}$$

The function $\mathbb{I}(A = a)$ is an *indicator function*; it takes the value 1 for people who receive treatment level a and 0 otherwise. The weighting term $p(A = a \mid Z)^{-1}$ corrects for treatment imbalance by assigning high weight to individuals who were *unlikely* to receive treatment a given their covariates, and low weight to those who were *likely* to receive it. This reweighting constructs a pseudo-population in which treatment is independent of Z, thereby emulating a randomized experiment. Finally, the empirical mean of the reweighted outcomes recovers the expected potential outcome $\mathbb{E}[Y^a]$. Units with $A \neq a$ contribute zero to this average.

We now show that the IPW functional is algebraically equivalent to the backdoor adjustment formula. At a high level, the proof uses the law of the unconscious statistician and the chain rule of probability to demonstrate that both estimators evaluate to the same target parameter.

Proof.

$$\begin{aligned}
\mathbb{E}\left[\frac{\mathbb{I}(A=a)\,Y}{p(A=a \mid Z)}\right] &= \sum_{Y,A,Z} p(Y,A,Z)\frac{\mathbb{I}(A=a)\,Y}{p(A=a \mid Z)} \\
&= \sum_{Y,Z} p(Y,A=a,Z)\frac{Y}{p(A=a \mid Z)} \\
&= \sum_{Y,Z} p(Y \mid A=a,Z)\,p(A=a \mid Z)\,p(Z)\frac{Y}{p(A=a \mid Z)} \\
&= \sum_{Y,Z} p(Y \mid A=a,Z)\,p(Z)\,Y \\
&= \sum_{Z} \mathbb{E}[Y \mid A=a,Z]\,p(Z).
\end{aligned}$$

□

2.4.3 Augmented Inverse Probability Weighting (AIPW)

The IPW estimator has several appealing features. In some settings, the propensity score $p(A \mid Z)$ is known by design—for example, in randomized clinical trials where treatment assignment probabilities are fixed by the study protocol. Even when the propensity score must be estimated, modeling $p(A \mid Z)$ is often relatively simple because A is frequently a binary variable.

However, IPW also has important limitations. When some propensity scores $p(A \mid Z)$ are very small, dividing by these small probabilities yields extremely large weights. This can cause numerical instability and inflate the variance of the estimator, resulting in wide confidence intervals. Such instability does not arise when using the backdoor adjustment estimator, which conditions directly on Z rather than reweighting observations.

A natural question, then, is whether we can combine the strengths of both approaches: the robustness of regression adjustment and the bias-correcting property of IPW. The *augmented inverse probability weighting* (AIPW) estimator accomplishes exactly this. By adding a regression-based correction term to the IPW estimator, AIPW blends the two approaches into a single estimator that is, importantly, *doubly robust*: it remains consistent for the ACE as long as either the propensity score model $p(A \mid Z)$ or the outcome regression model $\mathbb{E}[Y \mid A, Z]$ is correctly specified.

Let Z be a valid backdoor adjustment set relative to A and Y in a DAG $\mathcal{G}$. The *augmented inverse probability weighting* (AIPW) functional expresses the potential outcome mean $\mathbb{E}[Y^{(a)}]$ as

$$\mathbb{E}[Y^a] = \mathbb{E}\left[\frac{\mathbb{I}(A=a)}{p(A=a \mid Z)}\big(Y - \mathbb{E}[Y \mid A,Z]\big) \;+\; \mathbb{E}[Y \mid A=a,Z]\right].$$

This formulation makes explicit how AIPW enhances the IPW estimator by adding an outcome regression term $\mathbb{E}[Y \mid A, Z]$, thereby combining the strengths of reweighting and regression adjustment in a single expression.

Similarly, we can show that the AIPW functional is algebraically equivalent to the backdoor adjustment formula. The following derivation mirrors the argument used for the IPW functional,

relying on the law of the unconscious statistician and the chain rule of probability.

$$\mathbb{E}[Y^{a_1}] = \mathbb{E}\left[\frac{\mathbb{I}(A = a_1)}{p(A = a_1 \mid Z)}(Y - \mathbb{E}[Y \mid A, Z])\right] \; + \; \mathbb{E}\big[\mathbb{E}[Y \mid A = a_1, Z]\big].$$

By definition,

$$\mathbb{E}\big[\mathbb{E}[Y \mid A = a_1, Z]\big] = \sum_{z \in \mathcal{Z}} p(z)\, \mathbb{E}[Y \mid A = a_1, z].$$

It therefore suffices to show that

$$\mathbb{E}\left[\frac{\mathbb{I}(A = a_1)}{p(A = a_1 \mid Z)}\big(Y - \mathbb{E}[Y \mid A, Z]\big)\right] = 0.$$

Applying the law of unconscious statistician, we obtain

$$\begin{aligned}
&\mathbb{E}\left[\frac{\mathbb{I}(A = a_1)}{p(A = a_1 \mid Z)}(Y - \mathbb{E}[Y \mid A, Z])\right] \\
&= \sum_{Y,A,Z} p(Y, A, Z)\, \frac{\mathbb{I}(A = a_1)}{p(A = a_1 \mid Z)}\big\{Y - \mathbb{E}[Y \mid A, Z]\big\} \\
&= \sum_{Y,Z} p(Y, a_1, Z)\, \frac{Y - \mathbb{E}[Y \mid A = a_1, Z]}{p(A = a_1 \mid Z)} \\
&= \sum_{Y,Z} p(Z)\, p(A = a_1 \mid Z)\, p(Y \mid A = a_1, Z)\, \frac{Y - \mathbb{E}[Y \mid A = a_1, Z]}{p(A = a_1 \mid Z)} \\
&= \sum_{Z} p(Z) \sum_{Y} p(Y \mid A = a_1, Z)\, \big\{Y - \mathbb{E}[Y \mid A = a_1, Z]\big\} \\
&= \sum_{Z} p(Z) \left(\sum_{Y} p(Y \mid A = a_1, Z)\, Y \; - \; \mathbb{E}[Y \mid A = a_1, Z] \sum_{Y} p(y \mid A = a_1, Z)\right) \\
&= \sum_{Z} p(Z) \big(\mathbb{E}[Y \mid A = a_1, Z] \; - \; \mathbb{E}[Y \mid A = a_1, Z] \cdot 1\big) \\
&= 0.
\end{aligned}$$

Hence,

$$\mathbb{E}[Y^{a_1}] = \mathbb{E}\big[\mathbb{E}[Y \mid A = a_1, Z]\big] = \sum_{Z} p(Z)\, \mathbb{E}[Y \mid A = a_1, Z],$$

which is the backdoor adjustment formula for $\mathbb{E}[Y^{a_1}]$.

2.5 Selection Bias

Selection bias arises when inclusion into the analyzed sample depends systematically on variables related to the exposure, outcome, or both. A classic example is Berkson's bias, in which selection occurs by conditioning on a collider. Suppose we are interested in the relationship between a treatment A and an outcome Y, and both variables influence hospital admission R_S. Restricting analysis only to hospitalized patients ($R_S = 1$) induces a spurious association between A and Y, even if

they are independent in the target population. In this setting, the collider structure $A \to R_S \leftarrow Y$ generates bias through conditioning on the selection variable R_S.

A real-world epidemiologic example arises in studies of HIV-positive women receiving antiretroviral therapy in sub-Saharan Africa, where researchers may wish to estimate the effect of pregnancy on time to AIDS progression. If the study is conducted only among women attending antenatal care clinics, then both pregnancy and worsening HIV symptoms may influence clinic attendance. Conditioning on clinic attendance therefore induces a collider structure,

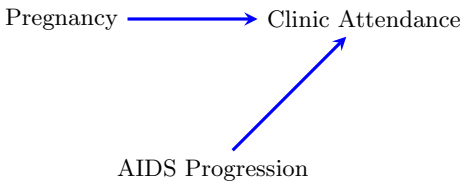


Figure 2.3: Berkson's bias through clinic-based selection.

which can create a non-causal association between pregnancy and AIDS progression, even if no such relationship exists in the target population.

However, selection bias can also arise without collider conditioning. For example, consider a study evaluating the effect of a job training program on post-training wages. Let A denote participation in the program, Y post-training earnings, and R_S an indicator of whether an individual responds to a follow-up wage survey. If individuals with higher realized wages are more likely to respond, then selection depends directly on the outcome through the mechanism $Y \to R_S$. This leads to overrepresentation of high-income individuals in the observed sample and distorts the selected distribution $p(Y, A \mid R_S = 1)$.

Under no confounding, the causal estimand satisfies

$$p(Y^a) = p(Y \mid A = a).$$

Yet this equality alone does not imply recoverability from the selected data, since the observed conditional distribution may itself be distorted by the selection mechanism. Our goal is therefore to characterize when the target causal quantity $p(Y^a)$ can still be recovered from data drawn under selection.

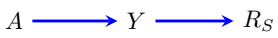


Figure 2.4: Outcome-dependent selection in the job training example.

An important extension arises when both confounding and selection bias are present simultaneously. In such settings, naive adjustment strategies generally fail, even in fully observed (Markovian) models. To address this problem, we rely on the results of Bareinboim and Tian (2015), which provide necessary and sufficient graphical and algorithmic conditions for the recoverability of causal effects under simultaneous selection and confounding bias. These results will play a central role in the theoretical developments and proofs presented in Chapter 5.

2.5.1 Recoverability in Markovian Models

In fully observed (Markovian) models, causal effects are identifiable from the observational distribution (Pearl, 2009). However, identifiability is not guaranteed when selection bias is present. The following result, due to Bareinboim and Tian (2015), characterizes when conditional distributions remain recoverable under selection bias.

Definition 1 (Recoverability from Selection Bias, Bareinboim and Tian (2015))**.** Let G_S be a causal graph over fully observed variables V, augmented with a selection node R_S, where $R_S = 1$ indicates inclusion in the observed dataset. The causal effect $p(Y^a)$ is said to be *recoverable from selection-biased data* in G_S if $p(Y^a)$ is uniquely determined from the observed selected distribution $p(V \mid R_S = 1)$ for all causal models compatible with G_S.

Theorem 1 (General Recoverability of Causal Effects, Bareinboim and Tian (2015))**.** *Let G_S be the causal graph augmented with the selection variable R_S, and let*

$$D = \mathrm{an}_{G_{V \setminus A}}(Y)$$

denote the ancestors of Y in a subgraph of G where the treatment A and edges incident to it are removed. The causal effect $p(Y^a)$ is recoverable from the selected distribution $p(V \mid R_S = 1)$ if and only if no variable in D is an ancestor of R_S in the original graph. When this condition holds,

$$p(Y^a) = \sum_{D \setminus Y} \prod_{V_i \in D} p(V_i \mid \mathrm{pa}_G(V_i), R_S = 1)\Big|_{A=a}.$$

A useful special case arises when the interventional distribution reduces to an observational conditional distribution.

Theorem 2 (Recoverability of Conditional Distributions, Bareinboim and Tian (2015))**.** *The conditional distribution $p(Y \mid A)$ is recoverable from selection-biased data if and only if*

$$Y \perp\!\!\!\perp R_S \mid A.$$

Corollary 2.1. *A causal effect $p(Y^a)$ is recoverable from selection-biased data if it can be reduced, via the rules of the do-calculus, to a conditional distribution $p(Y \mid A = a)$ that satisfies $Y \perp\!\!\!\perp R_S \mid A$.*

In the job training example, although there is no confounding and $p(Y^a) = p(Y \mid A = a)$, the condition $Y \perp\!\!\!\perp R_S \mid A$ fails due to outcome-dependent selection. Consequently, $p(Y \mid A = a)$ is not recoverable from $p(V \mid R_S = 1)$, and the causal effect $p(Y^a)$ cannot be consistently estimated from the observed data.

We next consider examples illustrating when recoverability succeeds or fails under different graphical structures.

Example 2.1 (Recoverability succeeds)**.** Consider the graph in Figure 2.5. Here, selection depends on treatment A and a pre-treatment covariate Z, while the outcome depends only on A. Since there

is no directed or backdoor path from R_S to Y after conditioning on A, we have

$$Y \perp\!\!\!\perp R_S \mid A.$$

Therefore, by the previous theorem, the conditional distribution $p(Y \mid A)$ is recoverable from the selected data, and by Corollary 1.1 the causal effect $p(Y^a)$ is also recoverable.

Intuitively, although selection depends on treatment assignment, once we condition on A, the selection mechanism carries no additional information about the outcome.

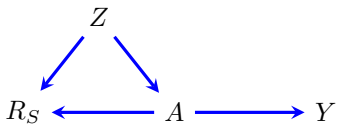


Figure 2.5: Recoverability succeeds since $Y \perp\!\!\!\perp R_S \mid A$.

Example 2.2 (Recoverability fails)**.** Now consider Figure 2.6, where an additional covariate W affects both treatment A and outcome Y. In this setting, conditioning only on A does not block the association between selection and outcome, since the path

$$R_S \leftarrow A \leftarrow W \rightarrow Y$$

remains active. Therefore,

$$Y \not\perp\!\!\!\perp R_S \mid A.$$

By the theorem, $p(Y \mid A)$ is not recoverable from the selected distribution.

Although the causal effect may still be identifiable in the full population after adjustment for W, Corollary 1.1 does not apply because the reduced observational quantity $p(Y \mid A)$ itself is not recoverable from the selected data.

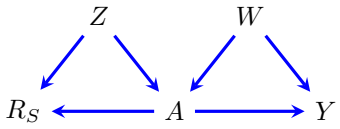


Figure 2.6: Recoverability fails since $Y \not\perp\!\!\!\perp R_S \mid A$.

2.6 Machine Learning

A popular definition of machine learning, due to Tom Mitchell, is as follows: A computer program is said to learn from experience E with respect to some class of tasks T, and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E. The most common form of machine learning is supervised learning. In this problem, the task T is to learn a mapping f from inputs $x \in X$ to outputs $y \in Y$. The inputs x are also called the features or predictors. In

this case, $X \in \mathbb{R}^p$, where p is the dimensionality of the vector (i.e., the number of input features). The output y is also known as the *label*, or *response variable*. The experience E is given in the form of a set of N input-output pairs $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, known as the *training set*, where n is called the *sample size*. In classification problems, the output space is a set of C unordered and mutually exclusive labels, known as *classes*, $Y = \{1, 2, \ldots, C\}$. If there are only two classes, often denoted by $y \in \{0, 1\}$, the problem is called *binary classification*. Now suppose that we want to predict a real-valued quantity $y \in \mathbb{R}$ instead of a class label $y \in \{1, \ldots, C\}$; this problem is known as *regression*. The performance measure P depends on the type of output being predicted.

We define the performance P of our model using a loss function $\mathcal{L} : \mathbb{R}^p \to \mathbb{R}$. We define $\mathcal{L}(y, f(x; \theta))$, which depends on θ for any given observation x and observed outcome y. The loss function $\mathcal{L}$ is defined so that its output is non-negative, and higher values represent worse predictors.

As currently defined, $\mathcal{L}$ allows us to evaluate the performance of different models, and so we now have each of the three pieces that Mitchell mentioned in his definition of machine learning. Selecting a good machine learning model can be thought of as solving the following optimization problem:

$$\theta^* = \arg\min_{\theta} \mathcal{L}(y, f(x; \theta)),$$

or in words, finding the optimal value of θ, a set of parameters that decides the complexity of the model, with respect to the given loss function $\mathcal{L}$.

2.7 Logistic Regression

Logistic regression is a supervised learning algorithm for binary classification. Given a feature vector $x \in \mathbb{R}^p$, the goal is to estimate the probability that an observation belongs to the positive class. Unlike linear regression, which produces unconstrained real-valued predictions, logistic regression maps its output to the unit interval $[0, 1]$, making predictions directly interpretable as probabilities.

Linear regression predicts a continuous outcome as a weighted sum of inputs,

$$\widehat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_d x_p, \tag{2.1}$$

but when the outcome is binary these predictions are not constrained to $[0, 1]$ and cannot be interpreted as probabilities. Logistic regression resolves this by passing the linear combination through the *sigmoid* function,

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \tag{2.2}$$

which is strictly increasing and maps any real number to the open interval $(0, 1)$. The predicted probability of the positive class is then

$$P(Y = 1 \mid x) = \sigma(\theta_0 + \theta_1 x_1 + \cdots + \theta_d x_p)\,. \tag{2.3}$$

The parameters $\theta = (\theta_0, \theta_1, \ldots, \theta_p)$ are estimated by maximizing the log-likelihood over n training

observations, which is equivalent to minimizing the binary cross-entropy loss,

$$\mathcal{L}(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\Big[y_i \log \hat{p}_i + (1-y_i)\log(1-\hat{p}_i)\Big], \tag{2.4}$$

where $\hat{p}_i = p(Y = 1 \mid x_i; \theta)$. Because this loss is convex in θ, gradient descent reliably converges to the global minimum. In this thesis, logistic regression is used to estimate the nuisance models required by causal estimators and serves as a baseline against which neural network nuisance models are compared in the simulation study of Chapter 5.

2.8 Neural Networks

A neural network takes an input vector of p variables $X = (X_1, X_2, \ldots, X_p)$ and builds a nonlinear function $f(X)$ to predict a response Y. The model learns this function by composing simple transformations across successive layers, each extracting increasingly abstract structure from the input.

A neural network with a single hidden layer of K neurons has the form

$$\begin{aligned} f(X) &= \beta_0 + \sum_{k=1}^{K} \beta_k h_k(X) \\ &= \beta_0 + \sum_{k=1}^{K} \beta_k \, g\left(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j\right), \end{aligned} \tag{2.5}$$

and is built up in two steps. In the first step, each of the K hidden units computes an *activation* A_k by applying a nonlinear function to a weighted sum of the inputs:

$$A_k = h_k(X) = g\left(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j\right), \quad k = 1, \ldots, K, \tag{2.6}$$

where $g(\cdot)$ is a nonlinear *activation function* chosen in advance. Each A_k can be interpreted as a learned feature — a distinct transformation of the raw inputs that captures some pattern in the data. In the second step, these activations are passed to the output layer, which forms a linear combination to produce the final prediction:

$$f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k. \tag{2.7}$$

All weights $w_{10}, \ldots, w_{Kp}$ and output coefficients $\beta_0, \ldots, \beta_K$ are estimated from the data by minimizing a suitable loss function, typically via gradient-based optimization. In early instances of neural networks, the *sigmoid* activation function was favored,

$$g(z) = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}},$$

which is the same function used in logistic regression to convert a linear function into probabilities between zero and one. The preferred choice in modern neural networks is the *ReLU* (*rectified linear unit*) activation function, which takes the form

$$g(z) = (z)_+ = \begin{cases} 0 & \text{if } z < 0, \\ z & \text{otherwise.} \end{cases}$$

A ReLU activation can be computed and stored more efficiently than a sigmoid activation. Although it thresholds at zero, because we apply it to a linear function the constant term w_{k0} will shift this inflection point. learning algorithms to handle imbalanced data more efficiently without altering combinations of X, and then squashes each through an activation function $g(\cdot)$ to transform it. The final model is linear in these derived variables. The name *neural network* originally came from thinking of these hidden units as analogous to neurons in the brain. We use neural networks to model our outcome regression and propensity score.

2.9 Class Imbalance

Class imbalance in binary classification tasks remains a significant challenge in machine learning, often resulting in poor performance on minority classes. Binary classification tasks frequently encounter imbalanced datasets, where one class significantly outnumbers the other. This imbalance can highly impact model performance, resulting in classifiers that excel at identifying the majority class but perform poorly on the minority class. In fields such as fraud detection, disease diagnosis, and rare event prediction, this bias can have serious consequences.

To address this challenge, researchers have developed various techniques targeting different stages of the machine learning pipeline to counteract the effects of class imbalance. These include data-level techniques, algorithm-level techniques, and integrated methods.

2.9.1 Data-level techniques

Data-level techniques involve modifying the dataset before the classifier is trained, including undersampling the majority class, oversampling the minority class, or creating synthetic data to increase the number of minority class.

Oversampling

Oversampling targets improving the representation of the minority class within a dataset, increasing its frequency to parity with the majority class. This adjustment can be achieved by simply duplicating existing instances or creating new, synthetic ones through methodologies like SMOTE (Synthetic Minority Oversampling Technique)(Chawla et al., 2002). SMOTE and its derivatives, such as Borderline-SMOTE and ADASYN, synthesize new samples by interpolating between minority class instances that connect via line segments to their nearest neighbors within the same class.

These approaches operate primarily in the feature space, thus injecting a higher degree of diversity into the lower class and supporting the model's ability to generalize from limited data.

The oversampling method offers several advantages in addressing imbalanced data, including increased model's generalization abilities by introducing a more comprehensive range of variability within the minority class and safeguards against the loss of essential information in minority class instances, which is particularly vital in datasets where each example holds significant value. However, oversampling can also have drawbacks, such as the potential for overfitting, increased computational complexity, the introduction of noise into the dataset, and limited information gain.

Undersampling

In contrast to oversampling, undersampling balances class distribution by reducing the size of the majority class, typically through the random removal of its instances. Undersampling offers several benefits: it can substantially reduce training time by shrinking the dataset, simplify the learning process, and lower the risk of majority-class bias, leading to more equitable model behavior.

However, undersampling also comes with notable drawbacks. Randomly discarding data risks eliminating informative or representative examples, potentially weakening the model's understanding of the majority class. Moreover, if the reduced dataset fails to capture the full variability within the majority class, the model may overfit, limiting its ability to generalize effectively.

2.9.2 Algorithm-level techniques

Algorithm-level techniques modify the learning algorithms to manage imbalanced data more efficiently without compromising the data itself, often through cost-sensitive learning, ensemble methods, or algorithm-specific adjustments. In this thesis, we focus on cost-sensitive learning. Cost-sensitive learning incorporates the idea that misclassification errors do not all carry equal consequences. In many real-world imbalanced classification tasks—such as fraud detection, medical diagnosis, or rare event prediction—incorrectly predicting the minority class (false negatives) is often far more costly than incorrectly predicting the majority class (false positives). Traditional classifiers treat all errors equally, leading to decision boundaries dominated by the majority class. Cost-sensitive learning addresses this by explicitly weighting errors according to their importance, thereby modifying the learning objective to penalize critical mistakes more heavily.

Two general strategies exist in cost-sensitive training:

1. Embedding costs directly into the loss function, where minority-class errors receive higher penalties,

2. Adjusting decision thresholds or margins, where classifiers shift the decision surface to prioritize the detection of the minority class.

Because cost-sensitive learning operates at the algorithmic level, it avoids modifying the dataset itself and maintains the original data distribution. This makes it particularly advantageous in applications where resampling may introduce noise, distort the feature space, or eliminate informative samples.

2.9.3 Hybrid Approaches

Hybrid methods combine data-level and algorithm-level strategies to handle class imbalance by leveraging the complementary strengths of both approaches. Instead of relying solely on resampling or solely on loss-function adjustments, hybrid techniques integrate these components to produce more stable and flexible models.

2.10 Evaluation Metrics for Imbalanced Classification

Evaluating models under class imbalance requires performance measures that account for the unequal importance of predicting the minority class. Traditional metrics such as overall accuracy can be misleading, as a classifier that always predicts the majority class may still achieve very high accuracy while performing poorly on the minority class. Most evaluation metrics are defined using the confusion matrix, consisting of

- True Positives (TP): correctly predicted positive instances,
- True Negatives (TN): correctly predicted negative instances,
- False Positives (FP): negative instances incorrectly labeled as positive,
- False Negatives (FN): negative instances incorrectly labeled as positive.

Accuracy

Accuracy is not suitable for imbalanced datasets because it is dominated by the majority class and does not reflect minority-class performance. A model may still achieve high overall accuracy even if its classification performance for minority categories is poor as long as it performs well in the majority categories. For example, if 99% of the testing data are negative samples, we can get a 99% accuracy by simply classifying all the testing data as a negative sample. So, the accuracy cannot be chosen as an evaluation index in imbalanced learning.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{2.8}$$

For the discussion that follows, we assume that the positive class is the minority class.

Precision

Precision measures the proportion of predicted positive instances that are truly positive. In other words, of all the cases the model flags as belonging to the minority class, precision tells us what fraction are correct. A model with low precision produces many false alarms — it labels majority-class instances as minority-class too often. This metric is most important in settings where acting on a false positive is costly; for example, flagging a healthy patient for an invasive procedure.

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{2.9}$$

Recall

Recall measures the proportion of actual positive instances that the model successfully identifies. Of all the true minority-class cases in the data, recall tells us what fraction the model catches. A model with low recall misses many minority-class instances, producing a high number of false negatives. Recall is most critical in settings where missing a positive case carries a high cost; for example, failing to detect a fraudulent transaction or a rare disease. Precision and recall are in tension: increasing the classification threshold tends to raise precision but lower recall, and vice versa.

$$\text{Recall} = \frac{TP}{TP + FN}. \tag{2.10}$$

F1 Score

Because precision and recall pull in opposite directions, it is useful to have a single metric that balances both. The F1 score is the harmonic mean of precision and recall, and it penalises extreme imbalance between the two. A model that achieves high precision but very low recall — or vice versa — will receive a low F1 score, whereas a model that performs reasonably well on both will score higher. The harmonic mean is preferred over the arithmetic mean here because it gives more weight to the lower of the two values.

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{2.11}$$

The F1 score ranges from 0 to 1, where 1 represents perfect precision and recall and 0 represents complete failure on at least one of the two.

ROC Curve and AUC

The ROC (Receiver Operating Characteristic) curve evaluates classifier performance across all possible classification thresholds, rather than at a single fixed cutoff. At each threshold, two quantities are computed: the True Positive Rate (TPR, identical to recall) and the False Positive Rate (FPR), defined as

$$\text{TPR} = \frac{TP}{TP + FN}, \qquad \text{FPR} = \frac{FP}{FP + TN}. \tag{2.12}$$

The ROC curve is obtained by plotting TPR against FPR as the threshold varies from 1 to 0. A perfect classifier passes through the top-left corner of the plot (TPR = 1, FPR = 0), while a classifier no better than random chance produces a diagonal line from $(0, 0)$ to $(1, 1)$. The Area Under the ROC Curve (AUC) summarises the entire curve as a single number. It can be interpreted as the probability that the model assigns a higher score to a randomly chosen positive instance than to a randomly chosen negative instance. An AUC of 1.0 indicates perfect discrimination; an AUC of 0.5 corresponds to random guessing. A higher AUC therefore indicates a model that ranks positive instances above negative ones more reliably, regardless of the chosen threshold.

Chapter 3

SMOTE

One of the most widely used methods for addressing class imbalance is the Synthetic Minority Over-sampling Technique (SMOTE), proposed by Chawla et al. (2002). Instead of duplicating existing minority samples as in traditional oversampling, SMOTE generates new synthetic data points by interpolating between real observations in feature space. Chawla et al. (2002) showed that the combination of SMOTE and undersampling performs better than plain undersampling alone. Since its introduction, the SMOTE paper has become one of the most cited works in the field of imbalanced learning, with over 35,000 citations. In this chapter, we describe how SMOTE works, examine its theoretical properties, and explore its implications when applied in causal inference settings, specifically during functional estimation.

3.1 How does SMOTE work?

Standard oversampling with replacement—that is, simply duplicating existing minority class instances—causes the learning algorithm to memorize the exact locations of those points rather than learning the broader region where minority class examples tend to occur, leading to overfitting rather than generalization. SMOTE addresses this by generating *synthetic* minority examples rather than copies, operating in feature space rather than data space.

For a given minority instance x_i, the algorithm identifies its k nearest minority class neighbors. A synthetic example is then created by selecting one of those neighbors, x_{nn}, and interpolating between them:

$$x_{\text{synthetic}} = x_i + \text{rand}(0, 1) \times (x_{nn} - x_i), \tag{3.1}$$

where $\text{rand}(0, 1)$ is a random scalar drawn uniformly from the unit interval. This places the synthetic point at a random position along the line segment connecting x_i to x_{nn} in feature space. By generating examples along these connecting segments rather than at the exact locations of existing points, SMOTE causes the classifier to learn broader and more general decision regions for the minority class, rather than fragmenting the feature space into small, highly specific pockets that are prone to overfitting.

Chawla et al. (2002) demonstrated that this approach improved recall and AUC performance across diverse datasets compared to both random oversampling and undersampling strategies. The improvement comes from the geometry of interpolation: minority class decision boundaries expand into the surrounding feature space rather than becoming more specific.

3.1.1 Algorithm

The full SMOTE procedure is summarized in Algorithm 3.1. The amount of oversampling is controlled by the parameter N, expressed as a percentage. For instance, $N = 200\%$ means two synthetic samples are generated per minority instance. The number of nearest neighbors k is a secondary hyperparameter; the original implementation uses $k = 5$ by default.

Algorithm 3.1 SMOTE

Input: Number of minority class samples T; oversampling amount $N\%$; number of nearest neighbors k

Output: $(N/100) \times T$ synthetic minority class samples

```
1:  if N < 100 then
2:      Randomize the T minority class samples
3:      T ← (N/100) × T
4:      N ← 100
5:  end if
6:  N ← ⌊N/100⌋
7:  for i ← 1 to T do
8:      Compute k nearest minority neighbors for sample i; store in nnarray
9:      POPULATE(N, i, nnarray)
10: end for

11: function POPULATE(N, i, nnarray)
12:     while N ≠ 0 do
13:         Choose a random index nn from {1, ..., k}
14:         for each feature attr do
15:             dif ← Sample[nnarray[nn]][attr] − Sample[i][attr]
16:             gap ← rand(0, 1)
17:             Synthetic[newindex][attr] ← Sample[i][attr] + gap × dif
18:         end for
19:         newindex ← newindex + 1
20:         N ← N − 1
21:     end while
22: end function
```

The key operation is in lines 11–15: for each feature, the algorithm computes the difference between the minority sample and its selected neighbor, scales it by a random gap, and adds it back. This is exactly equation (3.1) applied feature-by-feature. Because $gap \in (0, 1)$, every synthetic point lies strictly between the two real points in feature space.

3.2 Causal Implications of SMOTE

From a predictive modeling perspective, SMOTE is well-motivated and empirically effective. However, from a causal inference perspective, the implications are more complex and potentially harmful. The method changes the joint data-generating process $p(A, Z, Y)$ by introducing synthetic units that do not correspond to any real-world individual. This seemingly harmless modification has consequences for the validity of causal effect estimation.

Suppose treatment assignment is rare, i.e., $P(A = 1) \ll P(A = 0)$. To illustrate the effect of SMOTE under treatment imbalance, let (A_R, Z_R, Y_R) denote the real (observed) treatment, confounder, and outcome variables; let (A_{nn}, Z_{nn}, Y_{nn}) denote a nearest-neighbor observation drawn from the same (minority) treatment class; and let (A_S, Z_S, Y_S) denote the synthetic data generated by SMOTE. Under SMOTE, synthetic observations are generated by interpolating feature values from neighboring observations. Accordingly, the causal DAG contains arrows from Z_R and Z_{nn} into Z_S. The synthetic treatment label A_S, however, is not generated via interpolation—it is deterministically assigned to equal the minority class value by construction of the algorithm. Therefore, there is no causal edge from A_{nn} to A_S in the DAG.

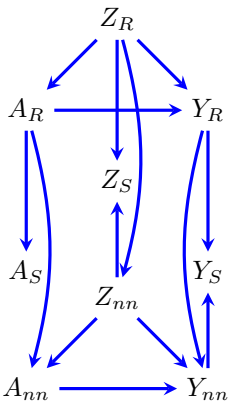


Figure 3.1: Causal diagram for SMOTE under treatment imbalance. Synthetic covariates Z_S are generated by interpolating from both the real observation and its nearest neighbor. The synthetic treatment label A_S is assigned deterministically, so no edge from A_{nn} to A_S exists.

Note that for rare outcomes, the causal DAG would look similar to Figure 3.1 but with a new edge from A_{nn} to A_S and without the edge from Y_{nn} to Y_S, reflecting the different interpolation structure when the outcome rather than the treatment is the minority class.

3.3 Distribution Shift after Applying SMOTE

To find the average causal estimand using the backdoor formula on the original (unresampled) dataset, we have

$$\mathbb{E}[Y^{(A=a)}] = \sum_{Z} \mathbb{E}_R[Y \mid A = a, Z]\ p_R(Z), \tag{3.2}$$

where $p_R(Z)$ denotes the marginal distribution of Z in the observed real data. After applying SMOTE, the marginal distribution of Z changes. Let $p_S(Z)$ denote the distribution of Z in the SMOTE-augmented dataset. Evaluating the same functional after resampling yields

$$\mathbb{E}[Y^{(A=a)}] = \sum_Z \mathbb{E}_S[Y \mid A = a, Z]\; p_S(Z), \tag{3.3}$$

which differs from (3.2) whenever $p_S(Z) \neq p_R(Z)$. As a result, the functional computed on the SMOTE-augmented data corresponds to a *different causal estimand* than the one defined on the original data. The following result formalizes this distributional shift.

Proof. For a univariate covariate Z, a synthetic value takes the form

$$Z_S = Z_i + \lambda(Z_{nn} - Z_i) = (1-\lambda)Z_i + \lambda Z_{nn},$$

where Z_i is a minority observation, Z_{nn} is a nearest neighbor, and $\lambda \sim \text{Unif}(0,1)$ with $\lambda \perp\!\!\!\perp Z_i, Z_{nn}$. We assume $Z_i, Z_{nn} \sim \mathcal{N}(0, \sigma^2)$.

Taking expectations,

$$\mathbb{E}[Z_S] = \mathbb{E}[(1-\lambda)Z_i + \lambda Z_{nn}] = \mathbb{E}[1-\lambda]\,\mathbb{E}[Z_i] + \mathbb{E}[\lambda]\,\mathbb{E}[Z_{nn}] = 0.$$

Thus $\mathbb{E}[Z_S] = \mathbb{E}[Z_i] = 0$, so the mean is preserved. However, the variance changes. Since $\mathbb{E}[Z_S] = 0$,

$$\text{Var}(Z_S) = \mathbb{E}[Z_S^2].$$

Expanding and using independence of λ and (Z_i, Z_{nn}):

$$\text{Var}(Z_S) = \mathbb{E}[(1-\lambda)^2]\,\mathbb{E}[Z_i^2] + \mathbb{E}[\lambda^2]\,\mathbb{E}[Z_{nn}^2] + 2\,\mathbb{E}[\lambda(1-\lambda)]\,\mathbb{E}[Z_i Z_{nn}].$$

Let $c = \mathbb{E}[Z_i Z_{nn}]$ capture the dependence between neighboring observations. Since $\lambda \sim \text{Unif}(0,1)$:

$$\mathbb{E}[\lambda] = \tfrac{1}{2}, \qquad \mathbb{E}[\lambda^2] = \tfrac{1}{3}, \qquad \mathbb{E}[(1-\lambda)^2] = \tfrac{1}{3}, \qquad \mathbb{E}[\lambda(1-\lambda)] = \tfrac{1}{6}.$$

Substituting:

$$\text{Var}(Z_S) = \sigma^2\big(\tfrac{1}{3}\big) + \sigma^2\big(\tfrac{1}{3}\big) + 2c\big(\tfrac{1}{6}\big) = \frac{2\sigma^2 + c}{3}.$$

In the original data, $\text{Var}(Z_i) = \sigma^2$. Except in the degenerate case where $c = \sigma^2$, the variance of Z_S differs from that of Z_i. Notably, since SMOTE selects nearest neighbors, Z_{nn} tends to be close to Z_i, so $c > 0$ in practice. However, $c < \sigma^2$ generically, which means $\text{Var}(Z_S) < \sigma^2$ — the synthetic distribution is *more concentrated* than the original. Although the synthetic values remain centered at the same location, their distribution does not coincide with the distribution from which the data were drawn, and $p_S(Z) \neq p_R(Z)$. This shift in distribution leads to a biased causal estimate. □

3.4 Simulation Counterexample for SMOTE

We generate data with a rare treatment and a known causal effect. For each unit,

$$\begin{aligned} Z &\sim \text{Bernoulli}(0.3), \\ A \mid Z &\sim \text{Bernoulli}\big(\sigma(-3.5 + 3Z)\big), \qquad \varepsilon \sim \mathcal{N}(0,1), \\ Y &= A + Z + 3AZ + \varepsilon, \end{aligned}$$

where errors are independent and $n = 100{,}000$. The treatment prevalence implied by this DGP is approximately 10%, creating a severe class imbalance. By construction, the true average causal effect is ACE = 1.9.

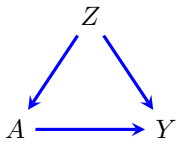


Figure 3.2: Causal diagram for the data-generating process. Z is a confounder affecting both treatment A and outcome Y. Conditioning on Z satisfies the backdoor criterion.

We apply SMOTE to balance the treatment classes before fitting the propensity score model, then estimate the ACE using the standard AIPW estimator. After applying SMOTE and recomputing the ACE, we obtain $\widehat{\text{ACE}} = 2.598$ with a 95% confidence interval (2.585, 2.611). Since $1.9 \notin (2.585, 2.611)$, SMOTE gives us a biased causal estimate in this example.

This counterexample shows that a procedure that improves predictive performance for the minority class — and is appropriate in a supervised learning context — can simultaneously destroy the validity of a downstream causal estimate. The SMOTE-augmented dataset has better class balance and higher F1 scores for the propensity model, yet it produces an estimate whose confidence interval does not even contain the true ACE.

Chapter 4

New reweighted AIPW Estimator

We show that the common resampling techniques used in machine learning don't preserve causal identifiability, and we propose a novel reweighted AIPW estimator to correct for the selection bias introduced by such resampling techniques.

4.1 Causal View of Resampling Methods

Let A denote the treatment, Z a confounder, and Y the outcome. Let V be the set of observed variables in the causal model, so that in our setting $V = \{Z, A, Y\}$. Let R_S denote the selection indicator, where $R_S = 1$ means that an observation is included in the resampled dataset.

For any graph G, let $\text{an}_G(X)$ denote the set of ancestors of variable X in G, and let $G_{V \setminus A}$ denote the subgraph graph obtained by removing A and all edges incident to A.

By Theorem 1, the causal effect $p(Y^a)$ is recoverable from the selected distribution $p(V \mid R_S = 1)$ if and only if no variable in $D = \text{an}_{G_{V \setminus A}}(Y)$ is also an ancestor of the selection variable R_S in the original graph G.

Outcome-dependent resampling (Rare Outcome)

In outcome-dependent resampling, the resampling depends only on the outcome variable Y, and therefore, the selection node R_S becomes a descendant of Y.

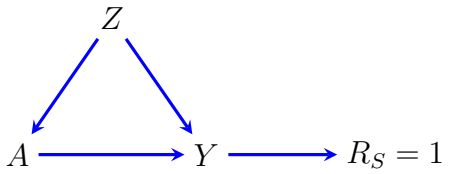


Figure 4.1: Causal diagram illustrating outcome-dependent resampling

Since Z is a common ancestor of the outcome Y and R_S, we have $\text{an}_{G_{V \setminus A}}(Y) \cap \text{an}_G(R_S) = \{Z\} \neq \emptyset$. Therefore, the causal effect $p(Y^a)$ is not recoverable under outcome-dependent resampling.

Treatment-Dependent Resampling (Rare Treatment)

When resampling depends on the treatment variable A, the selection node R_S is a descendant of A. However, Theorem 1 does not hold under treatment-dependent resampling because Y is an ancestor of R_S, violating the recoverability condition. In particular, after intervening on A, the relevant ancestral set is $D = \{Z, Y\}$, and since $Z \in \text{an}_G(R_S)$, the causal effect $p(Y^a)$ is also not recoverable in this setting.

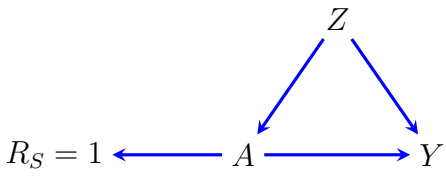


Figure 4.2: Causal diagram illustrating treatment-dependent resampling

Thus, any form of undersampling or oversampling that is either treatment-dependent or outcome-dependent will therefore yield a biased causal estimate due to selection bias.

Class Weighting

Another common strategy for handling class imbalance with binary data is class weighting, where the classification function is scaled to penalize misclassification asymmetrically. Class weighting assigns higher weights to the samples of the minority class and lower weights to the majority class during the training process.

Consider a binary treatment model fit via logistic regression with predicted probabilities $\hat{a}_i = p(A_i = 1 \mid Z_i; \theta)$. The (unweighted) negative log-likelihood is

$$\mathcal{L}(\theta) = -\sum_{i=1}^{n} \Big[a_i \log \hat{a}_i + (1 - a_i) \log(1 - \hat{a}_i) \Big]. \tag{4.1}$$

Introducing class weights w_1 for $A = 1$ and w_0 for $A = 0$ yields

$$\mathcal{L}_{\text{weighted}}(\theta) = -\sum_{i=1}^{n} \Big[w_1 \, a_i \log \hat{a}_i + w_0 \, (1 - a_i) \log(1 - \hat{a}_i) \Big], \tag{4.2}$$

with the balancing choice

$$w_1 = \frac{n}{2n_1}, \qquad w_0 = \frac{n}{2n_0},$$

where $n_1 = \sum_i a_i$, $n_0 = n - n_1$ and $w_0 + w_1 = 1$.

An equivalent way to understand class weighting is to view each observation as being *replicated* according to its assigned weight. Let W_i denote the class weight for observation i. Conceptually, we imagine forming a new dataset in which each observation appears W_i times. When W_i is not an integer, the replication is interpreted probabilistically: the observation is included an additional time with probability equal to the fractional part of W_i.

This stochastic–replication viewpoint makes clear that class weighting operates as a resampling procedure, not merely as a modification of the loss function.

Under this interpretation, weighting induces an implicit *selection variable* R_S that depends on the treatment A, since the probability that a unit is "replicated" or retained in the weighted sample is determined by its class membership and thus by A.

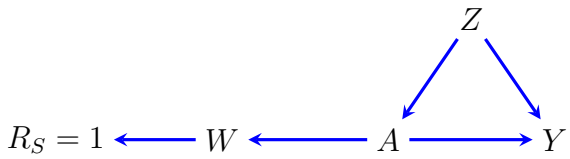


Figure 4.3: Causal diagram illustrating class-weighting

As shown above, Z is a common ancestor of both Y and R_S, violating the recoverability condition of Bareinboim and Tian (2015). As a result, the causal effect $p(Y^a)$ estimated from class-weighted data is also biased.

4.2 Reweighted AIPW Estimator for Rare Treatment

In many applications, treatment assignment is highly imbalanced. Although undersampling, oversampling, and class weighting are commonly used to "balance" the data, we have shown that these procedures introduce a selection bias and therefore applying standard AIPW to the resampled data gives us a biased ACE estimate.

To address this issue, we introduce a *reweighted augmented inverse probability weighting (rAIPW)* estimator that explicitly adjusts for this selection bias. The rAIPW framework enables practitioners to freely apply resampling or class-weighting techniques to handle imbalance in nuisance model estimation, while guaranteeing that the final estimator recovers the causal effect in the original target population.

Theorem 3 (Reweighted AIPW under treatment-dependent resampling)**.** *Let $R_S \in \{0,1\}$ denote the selection indicator after using a resampling procedure applied to an imbalanced dataset (Y, A, Z). We define a weight*

$$w(a, Z) = \frac{1}{p(R_S = 1 \mid Z)\, p(A = a \mid R_S = 1, Z)},$$

and the estimator

$$\widehat{\mathbb{E}}[Y^a] = \frac{1}{n} \sum_{i=1}^{n} \Big[w_i(a)\, \mathbb{I}(A_i = a) \big(Y_i - \widehat{\mathbb{E}}[Y \mid A_i, Z_i]\big) + \widehat{\mathbb{E}}[Y \mid A = a, Z_i] \Big].$$

accounts for the selection bias due to resampling, and the estimator recovers the true causal estimand

$$\mathbb{E}[Y^a].$$

4.2.1 Proof of rAIPW for rare treatment

In this section, we establish that the reweighted augmented inverse probability weighting (rAIPW) functional recovers the causal estimand $\mathbb{E}[Y^a]$ in the presence of treatment-dependent resampling. In particular, we show that the reweighting scheme removes the selection bias induced by such resampling procedures, and that the rAIPW functional reduces to the backdoor-adjusted estimand $\mathbb{E}_Z[\mathbb{E}(Y \mid A = a, Z)]$.

Proof.

$$\begin{aligned}\mathbb{E}[Y^a] &= \mathbb{E}\left[\frac{R_S \times \mathbb{I}(A=a)}{p(R_S=1 \mid Z)\, p(A=a \mid R_S=1, Z)} \times \big(Y - \mathbb{E}[Y \mid A, Z]\big) + \mathbb{E}[Y \mid A=a, Z]\right] \\ &= \mathbb{E}\left[\frac{R_S \times \mathbb{I}(A=a) \times Y}{p(R_S=1 \mid Z)\, p(A=a \mid R_S=1, Z)}\right] - \mathbb{E}\left[\frac{R_S \times \mathbb{I}(A=a) \times \mathbb{E}[Y \mid A, Z]}{p(R_S=1 \mid Z)\, p(A=a \mid R_S=1, Z)}\right] + \mathbb{E}[\mathbb{E}[Y \mid A=a, Z]]\,.\end{aligned}$$

To conclude that the rAIPW expression reduces to the backdoor estimand $\mathbb{E}_Z[\mathbb{E}(Y \mid A = a, Z)]$, it suffices to show that the augmentation term has expectation zero, i.e.

$$\mathbb{E}\left[\frac{R_S \times \mathbb{I}(A=a)}{p(R_S=1 \mid Z)\, p(A=a \mid R_S=1, Z)} \times \big(Y - \mathbb{E}[Y \mid A, Z]\big)\right] = 0.$$

$$\begin{aligned}&\mathbb{E}\left[\frac{R_S \times \mathbb{I}(A=a)}{p(R_S=1 \mid Z) \times p(A=a \mid R_S=1, Z)} \times \big(Y - \mathbb{E}[Y \mid A, Z]\big)\right] \\ &=^{(1)} \sum_{R_S, A, Z, Y} p(R_S, A, Z, Y) \times \frac{R_S \times \mathbb{I}(A=a) \times \big(Y - \mathbb{E}[Y \mid A, Z]\big)}{p(R_S=1 \mid Z) \times p(A=a \mid R_S=1, Z)} \\ &=^{(2)} \sum_{A, Z, Y} p(R_S=1, A, Z, Y) \times \frac{\mathbb{I}(A=a) \times \big(Y - \mathbb{E}[Y \mid A, Z]\big)}{p(R_S=1 \mid Z) \times p(A=a \mid R_S=1, Z)} \\ &=^{(3)} \sum_{A, Z, Y} p(R_S=1 \mid A, Z, Y)\, p(Y \mid A, Z)\, p(A \mid Z)\, p(Z) \times \frac{\mathbb{I}(A=a) \times \big(Y - \mathbb{E}[Y \mid A, Z]\big)}{p(R_S=1 \mid Z) \times p(A=a \mid R_S=1, Z)} \\ &=^{(4)} \sum_{A, Z, Y} p(Z)\, p(R_S=1 \mid Z)\, p(A \mid R_S=1, Z)\, p(Y \mid A, R_S=1, Z) \times \frac{\mathbb{I}(A=a) \times \big(Y - \mathbb{E}[Y \mid A, Z]\big)}{p(R_S=1 \mid Z) \times p(A=a \mid R_S=1, Z)} \\ &=^{(5)} \sum_{Z, Y} p(Z)\, p(Y \mid A=a, R_S=1, Z) \times \big(Y - \mathbb{E}[Y \mid A=a, Z]\big) \\ &=^{(6)} \sum_{Z, Y} p(Z)\, p(Y \mid A=a, Z) \times \big(Y - \mathbb{E}[Y \mid A=a, Z]\big) \quad (\text{by } Y \perp\!\!\!\perp R_S \mid (A, Z)) \\ &=^{(7)} \sum_{Z} p(Z) \sum_{Y} p(Y \mid A=a, Z) \times \big(Y - \mathbb{E}[Y \mid A=a, Z]\big) \\ &=^{(8)} \sum_{Z} p(Z) \Big(\mathbb{E}[Y \mid A=a, Z] - \mathbb{E}[Y \mid A=a, Z]\Big) = 0\end{aligned}$$

In (1) we apply the law of the unconscious statistician; in (2) we sum over R_S using the factor R_S in the numerator; in (3) we apply the chain rule of probability; in (4) we rewrite the joint as the

sampling distribution; (5) follows from cancellation of terms and the indicator $\mathbb{I}(A = a)$; in (6) we use the assumption $Y \perp\!\!\!\perp R_S \mid (A, Z)$; in (7) we recognize the inner sum as a conditional expectation; and in (8) the integrand evaluates to zero, yielding an overall expectation of zero.

Since the augmentation term is zero, the rAIPW estimand reduces to

$$\mathbb{E}[Y^a] \;=\; \mathbb{E}_Z\big[\mathbb{E}(Y \mid A = a, Z)\big].$$

□

4.2.2 Steps to Computing ACE using rAIPW in rare treatment

The procedure proceeds in three steps. First, we fit two logistic regression models: one for the probability of selection, $p(R_S = 1 \mid Z)$, using the original (pre-resampling) dataset, and another for the treatment mechanism within the resampled dataset, $p(A = 1 \mid R_S = 1, Z)$. We also estimate the outcome regression $\mathbb{E}[Y \mid A, Z]$ exactly the same way as we do when computing ACE using AIPW.

Next, for each observation, we compute a combined reweighting factor

$$w = \frac{1}{p(R_S = 1 \mid Z)\, p(A = a \mid R_S = 1, Z)}, \tag{4.3}$$

which adjusts for the selection bias.

Finally, the reweighted AIPW estimator for $\mathbb{E}[Y^a]$ is calculated as,

$$\widehat{\mathbb{E}}[Y^a] = \frac{1}{n}\sum_{i=1}^{n}\Big[w_i\,\mathbb{I}(A_i = a)\big(Y_i - \widehat{\mathbb{E}}[Y \mid A_i, Z_i]\big) + \widehat{\mathbb{E}}[Y \mid A = a, Z_i]\Big],$$

and the average causal estimate is obtained as the difference between the rAIPW estimates for $a = 1$ and $a = 0$.

4.3 Reweighted AIPW Estimator for Rare Outcome

Theorem 4 (Reweighted AIPW under outcome-dependent resampling)**.** *Let $R_S \in \{0, 1\}$ denote the selection indicator after using a resampling procedure applied to an imbalanced dataset (Y, A, Z). We define a weight*

$$w(a, Z) = \frac{1}{p(R_S = 1 \mid Y, Z)\, p(A = a \mid Z)},$$

and the estimator

$$\widehat{\mathbb{E}}[Y^a] = \frac{1}{n}\sum_{i=1}^{n}\Big[w_i(a)\,\mathbb{I}(A_i = a)\big(Y_i - \widehat{\mathbb{E}}[Y \mid A_i, Z_i]\big) + \widehat{\mathbb{E}}[Y \mid A = a, Z_i]\Big].$$

accounts for the selection bias due to resampling, and the estimator recovers the true causal estimand $\mathbb{E}[Y^a]$.

4.3.1 Proof of rAIPW for rare outcome

In this section, we establish that the reweighted augmented inverse probability weighting (rAIPW) estimator recovers the causal estimand $\mathbb{E}[Y^a]$ in the presence of outcome-dependent resampling. In particular, we show that the reweighting removes the selection bias introduced by resampling procedures, and that the rAIPW functional reduces to the backdoor-adjusted estimand $\mathbb{E}_Z[\mathbb{E}(Y \mid A = a, Z)]$.

Proof.

$$\begin{aligned}\mathbb{E}[Y^a] &= \mathbb{E}\left[\frac{R_S \times \mathbb{I}(A=a)}{p(R_S=1 \mid Y,Z)\, p(A=a \mid Z)} \times \big(Y - \mathbb{E}[Y \mid A,Z]\big) + \mathbb{E}[Y \mid A=a,Z]\right] \\ &= \mathbb{E}\left[\frac{R_S \times \mathbb{I}(A=a) \times Y}{p(R_S=1 \mid Y,Z)\, p(A=a \mid Z)}\right] - \mathbb{E}\left[\frac{R_S \times \mathbb{I}(A=a) \times \mathbb{E}[Y \mid A,Z]}{p(R_S=1 \mid Y,Z)\, p(A=a \mid Z)}\right] + \mathbb{E}[\mathbb{E}[Y \mid A=a,Z]]\,.\end{aligned}$$

To conclude that the rAIPW expression reduces to the backdoor estimand $\mathbb{E}_Z[\mathbb{E}(Y \mid A = a, Z)]$, it suffices to show that the augmentation term has expectation zero, i.e.

$$\mathbb{E}\left[\frac{R_S \times \mathbb{I}(A=a)}{p(R_S=1 \mid Y,Z)\, p(A=a \mid Z)} \times \big(Y - \mathbb{E}[Y \mid A,Z]\big)\right] = 0.$$

$$\begin{aligned}&\mathbb{E}\left[\frac{R_S \times \mathbb{I}(A=a)}{p(R_S=1 \mid Y,Z) \times p(A=a \mid Z)} \times \big(Y - \mathbb{E}[Y \mid A,Z]\big)\right] \\ &=^{(1)} \sum_{R_S,A,Z,Y} p(R_S,A,Z,Y) \times \frac{R_S \times \mathbb{I}(A=a) \times \big(Y - \mathbb{E}[Y \mid A,Z]\big)}{p(R_S=1 \mid Y,Z) \times p(A=a \mid Z)} \\ &=^{(2)} \sum_{A,Z,Y} p(R_S=1,A,Z,Y) \times \frac{\mathbb{I}(A=a) \times \big(Y - \mathbb{E}[Y \mid A,Z]\big)}{p(R_S=1 \mid Y,Z) \times p(A=a \mid Z)} \\ &=^{(3)} \sum_{A,Z,Y} p(R_S=1 \mid A,Z,Y)\, p(Y \mid A,Z)\, p(A \mid Z)\, p(Z) \times \frac{\mathbb{I}(A=a) \times \big(Y - \mathbb{E}[Y \mid A,Z]\big)}{p(R_S=1 \mid Y,Z) \times p(A=a \mid Z)} \\ &=^{(4)} \sum_{A,Z,Y} p(Z)\, p(R_S=1 \mid Y,Z)\, p(A \mid Z)\, p(Y \mid A,Z) \times \frac{\mathbb{I}(A=a) \times \big(Y - \mathbb{E}[Y \mid A,Z]\big)}{p(R_S=1 \mid Y,Z) \times p(A=a \mid Z)} \\ &=^{(5)} \sum_{Z,Y} p(Z)\, p(Y \mid A=a,Z) \times \big(Y - \mathbb{E}[Y \mid A=a,Z]\big) \\ &=^{(6)} \sum_{Z,Y} p(Z)\, p(Y \mid A=a,Z) \times \big(Y - \mathbb{E}[Y \mid A=a,Z]\big) \quad \text{(by cancellation)} \\ &=^{(7)} \sum_{Z} p(Z) \sum_{Y} p(Y \mid A=a,Z) \times \big(Y - \mathbb{E}[Y \mid A=a,Z]\big) \\ &=^{(8)} \sum_{Z} p(Z) \Big(\mathbb{E}[Y \mid A=a,Z] - \mathbb{E}[Y \mid A=a,Z]\Big) = 0.\end{aligned}$$

In (1) we apply the law of the unconscious statistician; in (2) we sum over R_S using the factor R_S in the numerator; in (3) we apply the chain rule of probability; in (4) we use the rare-outcome

sampling assumption $A \perp\!\!\!\perp R_S \mid (Y, Z)$ so that $p(R_S = 1 \mid A, Z, Y) = p(R_S = 1 \mid Y, Z)$ and we rewrite the remaining factors as $p(A \mid Z)$ and $p(Y \mid A, Z)$; (5) follows from cancellation of $p(R_S = 1 \mid Y, Z)$ and the indicator $\mathbb{I}(A = a)$ together with $p(A \mid Z)/p(A = a \mid Z)$; in (6) we simplify; in (7) we recognize the inner sum as a conditional expectation; and in (8) the integrand evaluates to zero, yielding an overall expectation of zero.

Since the augmentation term is zero, the rAIPW estimand reduces to

$$\mathbb{E}[Y^a] \;=\; \mathbb{E}_Z\big[\mathbb{E}(Y \mid A = a, Z)\big].$$

□

4.3.2 Steps to Computing ACE using rAIPW in rare outcomes

Similarly, we fit two logistic regression models: one for the probability of selection, $p(R_S = 1 \mid Y, Z)$, using the original (pre-resampling) dataset, and another for the treatment mechanism within the resampled dataset, $p(A = 1 \mid Z)$. We also estimate the outcome regression $\mathbb{E}[Y \mid A, Z]$ exactly the same way as we do when computing ACE using AIPW.

Next, for each observation, we compute a combined reweighting factor

$$w = \frac{1}{p(R_S = 1 \mid Y, Z)\, p(A = a \mid Z)}, \tag{4.4}$$

which adjusts for the selection bias.

Finally, the reweighted AIPW estimator for $\mathbb{E}[Y^a]$ is calculated as,

$$\widehat{\mathbb{E}}[Y^a] = \frac{1}{n} \sum_{i=1}^{n} \Big[w_i\, \mathbb{I}(A_i = a)\big(Y_i - \widehat{\mathbb{E}}[Y \mid A_i, Z_i]\big) + \widehat{\mathbb{E}}[Y \mid A = a, Z_i] \Big],$$

and the average causal estimate is obtained as the difference between the rAIPW estimates for $a = 1$ and $a = 0$.

4.4 Recovering Propensity Scores with the Odds Ratio Factorization

The odds ratio factorization of the propensity score, originally proposed by Chen (2007), allows us to identify joint probability $p(R_S, A \mid Z)$. An *odds ratio* between two variables is a statistic that measures the level of association between two variables X and Y. It is defined in terms of probabilities as

$$OR(X, Y \mid Z) = \frac{p(X \mid Y, Z)}{p(X = x_0 \mid Y, Z)} \times \frac{p(X = x_0 \mid Y = y_0, Z)}{p(X \mid Y = y_0, Z)}.$$

where x_0 is the *reference value* of X and y_0 is the *reference value* of Y. Reference values are

arbitrary values within the domain of X and Y that we choose to define the odds ratio over. From the above definition, we can see that the odds ratio is 1 whenever $X = x_0$ or $Y = y_0$ (either variable is at its reference value). Further, the odds ratio is 1 for *all* values of X, Y, Z if and only if $X \perp\!\!\!\perp Y \mid Z$. This characterizes all settings where the OR can attain a value of 1. The odds ratio is also symmetric, i.e., $OR(X, Y \mid Z) = OR(Y, X \mid Z)$. The range of the odds ratio function is $(0, \infty)$.

According to Chen (2007), the joint distribution $p(R_S, A \mid Z)$ can be represented using an odds-ratio factorization as

$$p(R_S, A \mid Z) = \frac{OR(R_S, A \mid Z)\, p(R_S = 1 \mid A, Z)\, p(A = 1 \mid R_S, Z)}{\mathcal{Z}(Z)}.$$

where the normalizing denominator $\mathcal{Z}(Z)$ is

$$\mathcal{Z}(Z) = \sum_{r \in \{0,1\}} \sum_{a \in \{0,1\}} OR(r, a \mid Z)\; p(R_S = 1 \mid A = a, Z)\; p(A = 1 \mid R_S = r, Z).$$

Here, the odds ratio function $OR(R_S, A \mid Z)$ captures the residual association between R_S and A after conditioning on Z, and is defined as

$$OR(R_S, A \mid Z) = \frac{p(R_S \mid A, Z)}{p(R_S = 1 \mid A, Z)} \times \frac{p(R_S = 1 \mid A = 1, Z)}{p(R_S \mid A = 1, Z)}.$$

This factorization decomposes the joint distribution into two components that are directly estimable from observed data—namely, $p(A \mid R_S = 1, Z)$ and $p(R_S \mid A = 1, Z)$—together with an odds ratio term that encodes the remaining dependence between A and R_S. We use the odds-ratio factorization for better interpretability and double robustness.

To see this, note that

$$p(R_S = 1 \mid Z) = \sum_{a \in \{0,1\}} p(R_S = 1, A = a \mid Z) \tag{4.5}$$

$$= p(R_S = 1, A = 0 \mid Z) + p(R_S = 1, A = 1 \mid Z). \tag{4.6}$$

Under the odds-ratio factorization,

$$\begin{aligned} p(R_S = 1 \mid Z) &= p(R_S = 1, A = 0 \mid Z) + p(R_S = 1, A = 1 \mid Z) \\ &= \frac{p(R_S = 1 \mid A = 1, Z)\, p(A = 0 \mid R_S = 1, Z)}{\sum_{r \in \{0,1\}} \sum_{a \in \{0,1\}} OR(r, a \mid Z)\, p(R_S = 1 \mid A = a, Z)\, p(A = 1 \mid R_S = r, Z)} \\ &\quad + \frac{p(R_S = 1 \mid A = 1, Z)\, p(A = 1 \mid R_S = 1, Z)}{\sum_{r \in \{0,1\}} \sum_{a \in \{0,1\}} OR(r, a \mid Z)\, p(R_S = 1 \mid A = a, Z)\, p(A = 1 \mid R_S = r, Z)}. \end{aligned}$$

To make the model estimable, we take $(A = 1, R_S = 1)$ as the reference cell, so the odds ratio equals one for every cell involving $A = 1$ or $R_S = 1$. The only nontrivial cell is $(A = 0, R_S = 0)$, for

which we introduce the parametric model

$$OR(A = 0, R_S = 0 \mid Z) = \exp(\eta_0 + \eta_1 Z),$$

with all other cells having $OR = 1$. Substituting into $\mathcal{Z}(Z)$ gives

$$\begin{aligned}\mathcal{Z}(\eta_0, \eta_1; Z) = &\underbrace{e^{\eta_0+\eta_1 Z} \cdot p(R_S{=}0 \mid A{=}0, Z) \cdot p(A{=}0 \mid R_S{=}0, Z)}_{\text{cell } (A{=}0,\ R_S{=}0)} \\ &+ \underbrace{p(R_S{=}1 \mid A{=}0, Z) \cdot p(A{=}0 \mid R_S{=}1, Z)}_{\text{cell } (A{=}0,\ R_S{=}1)} \\ &+ \underbrace{p(R_S{=}0 \mid A{=}1, Z) \cdot p(A{=}1 \mid R_S{=}0, Z)}_{\text{cell } (A{=}1,\ R_S{=}0)} \\ &+ \underbrace{p(R_S{=}1 \mid A{=}1, Z) \cdot p(A{=}1 \mid R_S{=}1, Z)}_{\text{cell } (A{=}1,\ R_S{=}1)}.\end{aligned}$$

The parameters (η_0, η_1) are then estimated by maximizing the log-likelihood of the observed (A, R_S) pairs:

$$\ell(\eta_0, \eta_1) = \sum_{i=1}^{n} \Big[\log \tilde{p}(R_{S,i}, A_i \mid Z_i;\, \eta_0, \eta_1) - \log \mathcal{Z}(\eta_0, \eta_1; Z_i)\Big],$$

where $\tilde{p}(R_S, A \mid Z;\, \eta_0, \eta_1)$ denotes the unnormalized numerator of the factorization evaluated at the observed cell for unit i. Once $(\hat{\eta}_0, \hat{\eta}_1)$ have been obtained, the marginal selection probability follows directly:

$$p(R_S = 1 \mid Z) = p(A = 0,\, R_S = 1 \mid Z) + p(A = 1,\, R_S = 1 \mid Z),$$

which yields the inverse selection weight required by the rAIPW estimator while correctly accounting for the treatment-dependent structure of the resampling mechanism.

Chapter 5

Simulation Study

This chapter evaluates the performance of the rAIPW estimator introduced in Chapter 4 through a controlled simulation study. Because real-world observational datasets rarely provide access to a known ground-truth causal effect, we use a fully synthetic data-generating process (DGP) where the true average causal effect (ACE) is known by construction. This allows us to assess not only whether our estimator produces a reasonable point estimate on a single dataset, but whether it behaves correctly *across many repeated draws* from the same DGP.

5.1 Data-Generating Process

The underlying causal structure is shown in Figure 5.1. The binary covariate Z simultaneously affects treatment assignment A and outcome Y, creating a standard backdoor confounding path. Conditioning on Z therefore satisfies the backdoor criterion and is sufficient to identify the causal effect of A on Y.

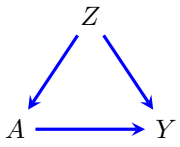


Figure 5.1: Causal diagram for the simulation study. Z confounds the effect of treatment A on outcome Y.

Data are generated according to the following structural equations:

$$\begin{aligned} Z &\sim \text{Bernoulli}(0.3), \\ A \mid Z &\sim \text{Bernoulli}\big(\sigma(-3.5 + 3Z)\big), \\ Y &= A + Z + 3AZ + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, 1), \end{aligned}$$

where $\sigma(x) = (1 + e^{-x})^{-1}$ is the logistic function and all noise terms are mutually independent.

The treatment assignment is deliberately imbalanced: the marginal probability of treatment is approximately 12%, producing a rare-treatment setting typical of applications in which resampling is routinely applied. By construction, the true ACE is

$$\text{ACE} = \mathbb{E}\Big[Y^{(1)} - Y^{(0)}\Big] = 1.9.$$

5.2 Estimators and Evaluation Protocol

We compare two estimators throughout the study.

Standard AIPW. The augmented inverse probability weighted estimator is fitted directly on the resampled dataset, with no correction for the sampling mechanism. This mirrors the practice of a practitioner who applies resampling to address class imbalance and then proceeds with a standard causal estimator, unaware that resampling has changed the effective target estimand.

Reweighted AIPW (rAIPW) for Rare Treatment. Our proposed estimator restores the original causal estimand after resampling has been applied. More details of the construction are given in Chapter 4.

The evaluation proceeds in two stages.

1. **Point estimation under a fixed sample size.** We fix $n = 2{,}000$ and apply each of four resampling strategies (undersampling, oversampling, class weighting, and SMOTE) before running the standard AIPW estimator. For undersampling, oversampling, and class weighting—where the resampling mechanism is fully characterised by a selection indicator R_s — we additionally run the rAIPW estimator. SMOTE is excluded from this comparison because it generates synthetic observations that do not correspond to any unit in the original dataset, making it impossible to define a valid selection probability $p(R_s = 1 \mid Z)$; the rAIPW correction therefore does not apply. For each combination, we report the estimated ACE, a 95% bootstrap confidence interval, and the F1 score of the fitted treatment model.

2. **Convergence across sample sizes.** We vary n over (100, 200, 500, 1,000, 2,000, 3,000) and repeat each experiment 200 times per sample size with independent random seeds. At each n we collect the distribution of estimates from both the undersampling and oversampling methods visualized as side-by-side box plots. This stage shows whether any remaining bias is present regardless of n or whether the bias disappears as the data accumulate.

5.3 Results

5.3.1 Point Estimates Under Fixed Sample Size

Table 5.1 summarizes the results for $n = 2{,}000$. As shown, the standard AIPW estimator is biased under every resampling strategy: across undersampling, oversampling, and class weighting, it con-

sistently returns estimates near 2.6, roughly 36% above the true ACE of 1.9. This is not numerical noise; it is a direct consequence of the estimand shift induced by resampling, as established theoretically in Chapter 4. However, the rAIPW estimator recovers accurate estimates in all cases where a correction is possible, with confidence intervals that cover the true value.

The F1 score column is instructive for a different reason. It does not track causal accuracy: undersampling achieves F1 = 0.85 yet the standard AIPW estimate is highly biased; oversampling achieves F1 = 0.82 with an equally biased estimate. This confirms the central message of Chapter 1 — a well-performing treatment model is neither necessary nor sufficient for unbiased causal estimation.

Resampling Strategy	**Estimator**	**Target**†	**Logistic Regression**		**Neural Network**	
			$\widehat{\text{ACE}}$ (95% CI)	**F1**	$\widehat{\text{ACE}}$ (95% CI)	**F1**
Baseline (Unbalanced)	Standard AIPW	✓	1.89 (1.64, 2.09)	—	1.89 (1.64, 2.10)	—
Undersampling	Standard AIPW	×	2.44 (2.14, 2.72)	0.85	2.44 (2.14, 2.72)	0.85
	Reweighted AIPW	✓	**1.77** (1.63, 2.17)	0.85	**1.77** (1.64, 2.18)	0.85
Oversampling	Standard AIPW	×	2.59 (2.57, 2.60)	0.82	2.58 (2.56, 2.60)	0.82
	Reweighted AIPW	✓	**1.90** (1.899, 1.901)	0.82	**1.90** (1.898, 1.902)	0.82
Class Weighting	Standard AIPW	×	2.60 (2.581, 2.615)	0.63	2.60 (2.581, 2.615)	0.63
	Reweighted AIPW	✓	**1.90** (1.865, 1.926)	0.63	**1.90** (1.866, 1.928)	0.63
SMOTE	Standard AIPW	×	2.60 (2.585, 2.611)	—	2.59 (2.584, 2.613)	—
	Reweighted AIPW	✓	—	—	—	—

† ✓ indicates the estimator targets the correct causal estimand (ACE = 1.9); × indicates it does not. Bold $\widehat{\text{ACE}}$ values highlight Reweighted AIPW estimates. Treatment prevalence is approximately 10% in all resampled settings. Dashes (—) denote results pending.

Table 5.1: Point estimates and 95% bootstrap confidence intervals for the ACE under four resampling strategies, two estimators, and two nuisance model classes. Fixed sample size $n = 2{,}000$; true ACE = 1.9.

5.3.2 Convergence Across Sample Sizes

Figures 5.2 and 5.3 display the distribution of estimates across 200 independent replicates for each sample size, under undersampling and oversampling respectively. Each box summarises 200 independently drawn datasets of size n; the red dashed line marks the true ACE of 1.9.

The standard AIPW estimates (left panel, salmon) sit persistently around 2.6 across *all* sample sizes. The interquartile range narrows as n grows — consistent with reduced sampling variance — but the median never approaches the true value. This is the hallmark of a *biased and inconsistent* estimator: more data does not help, because the estimator is targeting the wrong quantity.

The rAIPW estimates (right panel, blue) behave very differently. At $n = 100$ the distribution is wide and occasionally far from 1.9, reflecting the difficulty of estimating two nuisance models on a small resampled dataset. However, as n increases the boxes tighten and centre squarely on the red line. By $n = 1{,}000$ the median is indistinguishable from the true ACE in both scenarios,

and by $n = 2{,}000$–$3{,}000$ the interquartile range is narrow enough to be practically negligible. This convergence pattern is the empirical signature of a *consistent* estimator: given sufficient data, it reliably recovers the truth.

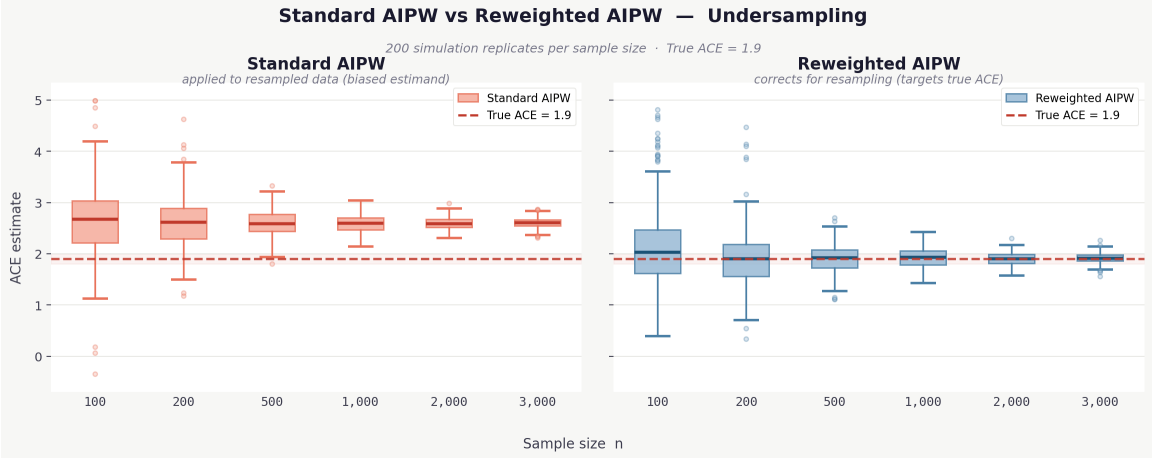


Figure 5.2: Distribution of ACE estimates across 200 simulation replicates per sample size under **undersampling**. Left: standard AIPW applied to the balanced subset. Right: rAIPW applied to the full dataset with resampling weights. The dashed line marks the true ACE = 1.9.

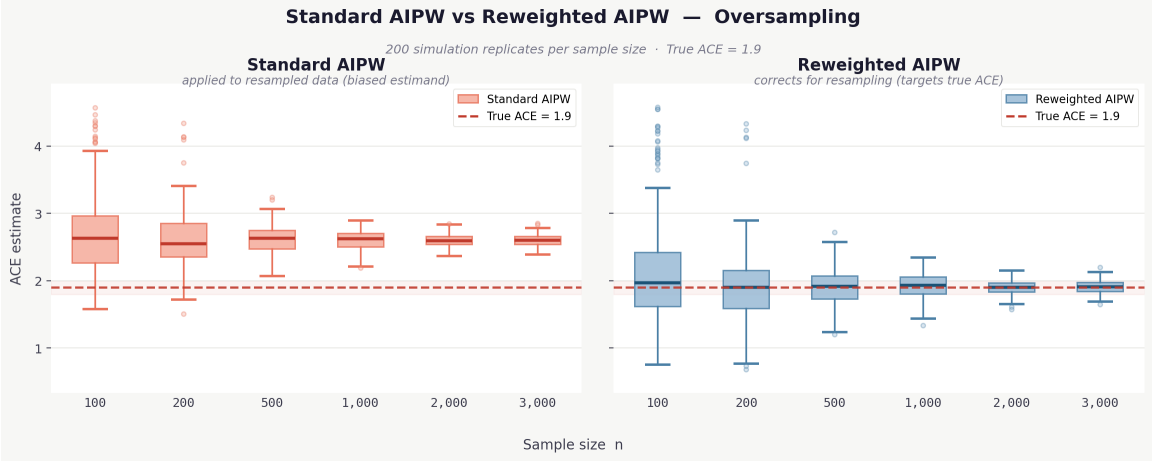


Figure 5.3: Distribution of ACE estimates across 200 simulation replicates per sample size under **oversampling**. Left: standard AIPW applied to the balanced dataset. Right: rAIPW applied to the full dataset with resampling weights. The dashed line marks the true ACE = 1.9.

5.4 Summary

Using the simulation, we have shown applying any resampling strategy prior to nuisance model estimation causes the standard AIPW estimator to target a shifted estimand, producing bias that does not diminish with sample size. Undersampling, oversampling, and class weighting are all prone

to selection bias in the causal estimate while SMOTE shifts the covariate distribution and thus leading to biased estimate.

The rAIPW estimator corrects for the selection bias by incorporating resampling probabilities into the inverse probability weights. Across all strategies and both nuisance model classes, it recovers estimates close to the true ACE of 1.9, with confidence intervals covering the true value. The convergence plots confirm that this is not a finite-sample coincidence: the estimator is consistent, with variance shrinking to zero as $n \to \infty$ while the median tracks the truth throughout.

These results provide strong empirical support for the theoretical claims of Chapter 4 and underscore the core argument of this thesis: predictive performance metrics such as the F1 score and causal validity are distinct objectives, and optimizing the former can actively undermine the latter.

Chapter 6

Discussion and Conclusion

The central contribution of this thesis is to investigate the applications of machine learning methods to address class imbalance in nuisance model estimation in observational causal inference studies. Through both theoretical and empirical investigations, we showed that undersampling, oversampling, and class weighting are all prone to selection bias and therefore yield biased causal estimates. We also showed that applying SMOTE to an imbalanced dataset leads to a distributional shift in our covariates, as evidenced by changes in their variances. Based on these results, we proposed and demonstrated a new method to correct for selection bias, called the reweighted Augmented Inverse Probability Weighting (rAIPW) estimator.

The main theoretical insight here is that resampling is not a neutral pre-processing step. When examined through the framework of causal graphs, each resampling strategy (except SMOTE) corresponds to an implicit selection variable, denoted as R_S. The inclusion of this variable depends on the treatment, the outcome, or both. As established by Bareinboim and Tian (2015), recovering the causal effect from such a selected distribution requires that no ancestor of the outcome in the altered graph is also an ancestor of the selection variable. Unfortunately, all standard resampling strategies violate this condition. Specifically, the covariate Z, which influences both the treatment A and the outcome Y, becomes an ancestor of R_S through either A or Y. Thus, the estimand calculated from the resampled data represents a different causal quantity than the one defined in the original population.

The rAIPW estimator addresses both treatment- and outcome-dependent selection bias by reweighting the standard augmented inverse probability weighting estimator using explicit selection probabilities. In particular, it incorporates $p(R_S = 1 \mid Z)$ and $p(R_S = 1 \mid Y, Z)$—corresponding to treatment-dependent and outcome-dependent resampling, respectively—directly into the inverse probability weights to correct for the non-random inclusion of observations into the resampled data.

6.1 Recommendations for Class Imbalance

The findings of this thesis suggest several concrete guidelines for practitioners working at the intersection of machine learning and causal inference. First, when propensity score models are estimated on

data that has been resampled or reweighted for class balance, the resulting propensity scores should not be treated as estimates of $p(A \mid Z)$ in the original population. Second, predictive performance metrics such as F1 score, AUC, and balanced accuracy are not reliable proxies for the quality of nuisance estimates in a causal pipeline; a model that achieves better classification performance may produce worse causal estimates if it does so by distorting the treatment mechanism. Third, when resampling is deemed necessary—for example, to stabilize numerical optimization during propensity score model training—the rAIPW estimator provides a principled way to correct for the resulting selection bias, provided that the selection probabilities can be estimated.

6.2 Limitations

Several limitations of the present work deserve acknowledgement. First, the theoretical guarantees established in Chapter 4 rely on correct specification of the selection mechanism. The rAIPW estimator requires consistent estimation of $p(R_S = 1 \mid Z)$, and misspecification of this nuisance model introduces its own bias that is not corrected by the reweighting procedure. In practice, this places a meaningful burden on the analyst to carefully model the resampling mechanism, which may not always be transparent or documented in applied settings.

Second, the compound inverse probability weights in the rAIPW estimator can exhibit high variance in finite samples, as the simulation results for small n illustrate. When either $p(R_S = 1 \mid Z)$ or $p(A = 1 \mid R_S = 1, Z)$ is close to zero for some units, the resulting weights become extreme and destabilize estimation. Practical remedies such as weight trimming, normalization, or cross-fitting were not investigated here and remain an open consideration for applied use.

Third, the framework developed in this thesis is restricted to binary treatments and binary outcomes. While this setting naturally captures the class imbalance problem as it is most commonly encountered in the machine learning literature, many real-world causal questions involve continuous, multi-valued, or time-varying treatments. The rAIPW estimator as formulated does not extend directly to these cases.

Finally, the rAIPW correction does not apply to SMOTE. Unlike undersampling, oversampling, and class weighting — which preserve the original covariate distribution and select or replicate existing units — SMOTE generates synthetic observations by interpolating between real data points. Because these synthetic units do not correspond to any unit in the target population, the selection probability $p(R_S = 1 \mid Z)$ is not well-defined for them, and the reweighting strategy that underpins the rAIPW estimator cannot be applied.

6.3 Future Work

The limitations outlined above point naturally to several directions for future research. The most immediate extension is to move beyond binary treatments. The current framework formulates class imbalance as a rare-event problem in which one binary treatment arm is substantially underrepresented, but analogous imbalance issues arise whenever treatment takes multiple discrete values or

lies on a continuous scale. Extending the rAIPW framework to these settings would require generalizing the selection-bias analysis beyond binary R_S indicators and developing reweighting strategies compatible with generalized propensity score models.

A second direction concerns variance reduction for the rAIPW estimator. The simulation results show that estimation is reliable at moderate to large sample sizes, but can be unstable for small n, where extreme weights inflate variance. Cross-fitting — in which nuisance models are trained on held-out folds to reduce overfitting bias — and weight stabilization or trimming strategies are natural candidates for investigation. Understanding the efficiency properties of the rAIPW estimator and whether a semiparametrically efficient version can be constructed would further strengthen its theoretical foundations.

Perhaps the most technically challenging extension is to develop a principled analog of the rAIPW correction for SMOTE. Because SMOTE modifies the marginal covariate distribution $p(Z)$ by introducing synthetic interpolated points, the target estimand is itself shifted relative to the original population: inference is effectively performed under a covariate distribution that does not correspond to any real data-generating process. A correction for SMOTE would therefore need to account not only for selection bias but also for this distributional shift, requiring tools from covariate shift and domain adaptation in addition to the selection-bias framework developed here. Whether such a correction can be made to target the original population estimand — or whether SMOTE-augmented datasets are fundamentally incompatible with population-level causal inference — remains an open question.

Appendix A

Additional Simulation Details

A.1 Additional Details regarding Simulations in Section 1.1

Define the function $\sigma(x)$ as

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

The simulations were performed under the following data-generating process:

$$\begin{aligned}
&W_1,\, W_2 \sim \mathcal{N}(0, 1),\\
&A \sim \text{Bernoulli}(\sigma(-2 + 1.2\, W_1 + 1.5\, W_2))\,,\\
&O_1 = 0.5\, W_1 + \varepsilon_1, \qquad O_2 = 0.5\, W_2 + \varepsilon_2, \qquad O_3 = \varepsilon_3,\\
&Y = 2\, A + 2\, O_1 - 1.2\, O_2 + 1.5\, O_3 + \varepsilon_Y,
\end{aligned}$$

where all $\varepsilon \sim \mathcal{N}(0, 1)$ are mutually independent and the sample size is $n = 5{,}000$.

A.2 Details regarding Simulations in Section 1.2

Define the function $\sigma(x)$ as

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

The simulations in the motivating example were performed under the following data-generating process:

$$\begin{aligned}
&W_1,\, W_2 \sim \mathcal{N}(0, 1),\\
&A \sim \text{Bernoulli}(\sigma(-2 + 1.2\, W_1 + 1.5\, W_2))\,,\\
&O_1 = 0.5\, W_1 + \varepsilon_1, \qquad O_2 = 0.5\, W_2 + \varepsilon_2, \qquad O_3 = \varepsilon_3,\\
&Y = 2\, A + 2\, O_1 - 1.2\, O_2 + 1.5\, O_3 + \varepsilon_Y,
\end{aligned}$$

where all $\varepsilon \sim \mathcal{N}(0,1)$ are mutually independent and the sample size is $n = 5{,}000$.

Bibliography

Balzer, L., Ahern, J., Galea, S., and van der Laan, M. (2016). Estimating effects with rare outcomes and high dimensional covariates: Knowledge is power. *Epidemiologic Methods*, 5(1):1–18.

Bang, H. and Robins, J. M. (2005). Doubly robust estimation in missing data and causal inference models. *Biometrics*, 61(4):962–973.

Bareinboim, E. and Tian, J. (2015). Recovering causal effects from selection bias. *Proceedings of the AAAI Conference on Artificial Intelligence*, 29(1).

Bhattacharya, R. and Nabi, R. (2022). On testability of the front-door model via verma constraints. In *Proceedings of the Thirty-Eighth Conference on Uncertainty in Artificial Intelligence*, pages 202–212. PMLR.

Bhattacharya, R., Nabi, R., and Shpitser, I. (2022). Semiparametric inference for causal effects in graphical models with hidden variables. *Journal of Machine Learning Research*, 23(295):1–76.

Chawla, N. V., Bowyer, K. W., Hall, L. O., and Kegelmeyer, W. P. (2002). Smote: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357.

Chen, H. Y. (2007). A semiparametric odds ratio model for measuring association. *Biometrics*, 63(2):413–421.

Entner, D., Hoyer, P., and Spirtes, P. (2013). Data-driven covariate selection for nonparametric estimation of causal effects. In *Artificial Intelligence and Statistics*, pages 256–264. PMLR.

Hansen, B. E. (2000). Sample splitting and threshold estimation. *Econometrica*, 68(3):575–603.

Hernán, M. A., Wang, W., and Leaf, D. E. (2022). Target trial emulation: A framework for causal inference from observational data. *JAMA*, 328(24):2446–2447.

Horvitz, D. G. and Thompson, D. J. (1952). A generalization of sampling without replacement from a finite universe. *Journal of the American statistical Association*, 47(260):663–685.

Imbens, G. W. (2020). Potential outcome and directed acyclic graph approaches to causality: Relevance for empirical practice in economics. *Journal of Economic Literature*, 58(4):1129–1179.

Keith, K. A., Feldman, S., Jurgens, D., Bragg, J., and Bhattacharya, R. (2023). Rct rejection sampling for causal estimation evaluation. *arXiv preprint arXiv:2307.15176*.

Koller, D. and Friedman, N. (2009). *Probabilistic Graphical Models: Principles and Techniques*. MIT Press.

Malinsky, D., Shpitser, I., and Richardson, T. (2019). A potential outcomes calculus for identifying conditional path-specific effects. In *The 22nd International Conference on Artificial Intelligence and Statistics*, pages 3080–3088. PMLR.

Newey, W. K. (1990). Semiparametric efficiency bounds. *Journal of Applied Econometrics*, 5(2):99–135.

Newey, W. K. and Robins, J. M. (2018). Cross-fitting and fast remainder rates for semiparametric estimation. *arXiv preprint arXiv:1801.09138*.

Neyman, J. (1923). On the application of probability theory to agricultural experiments: Essay on principles. *Annals of Agricultural Sciences*, pages 1–51.

Pearl, J. (1995). Causal diagrams for empirical research. *Biometrika*, 82(4):669–710.

Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*. Cambridge University Press.

Richardson, T. S. and Robins, J. M. (2013). Single world intervention graphs (SWIGs): A unification of the counterfactual and graphical approaches to causality. *Center for the Statistics and the Social Sciences, University of Washington Working Paper 128*, pages 1–146.

Robins, J. (1986). A new approach to causal inference in mortality studies with a sustained exposure period: Application to control of the healthy worker survivor effect. *Mathematical Modelling*, 7(9–12):1393–1512.

Spirtes, P., Glymour, C., and Scheines, R. (2001). *Causation, Prediction, and Search*. MIT Press.

Tamarchenko, E. (2023). Combining optimal adjustment set selection and post selection inference in unknown causal graphs.

Tchetgen Tchetgen, E. J., Robins, J. M., and Rotnitzky, A. (2010). On doubly robust estimation in a semiparametric odds ratio model. *Biometrika*, 97(1):171–180.

Tsiatis, A. A. (2006). *Semiparametric Theory and Missing Data*, volume 4. Springer.

van der Vaart, A. W. (2000). *Asymptotic Statistics*. Cambridge University Press.

Verma, T. and Pearl, J. (1990). Equivalence and synthesis of causal models. In *Proceedings of the Sixth Annual Conference on Uncertainty in Artificial Intelligence*, pages 255–270.

Yang, J., Bhattacharya, R., Lee, Y., and Westling, T. (2024). Statistical and causal robustness for causal null hypothesis tests. In *Proceedings of the Fortieth Conference on Uncertainty in Artificial Intelligence*.